Inventory Reconciliation Agent

System user guide: the exception worklist and the source simulator

Bimbo Bakeries USA · Lehigh Valley bakery (plant UBD) · demo sales day Saturday 12 September 2026

Version 0.5.0 · written from a full user test run on 10 November 2026

What is in this guide

How to use every part of the application, step by step, with screenshots taken during a full test run: signing in, the worklist, working as a team, decisions, sending corrections back to the source systems, the source simulator, integration faults, all fourteen scenarios (ten for the rules, four for the ML layer), the ML layer's anomaly detection, phones and tablets, workbook mode, the command line and the API.

The test report at the end lists every check that was run: 78 hands-on checks (75 passed) and the 141 automated tests (all passed), with the issues found.

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1. What the application does

Five systems at the bakery each hold part of the same story: how many buns customers ordered, how many are forecast, how many are already in the warehouse, and how many the plant must make. When one system misses a change, the plant makes too many or too few. The **Inventory Reconciliation Agent** checks that every system carries its part of the story, raises a **flag** when one does not, ranks the flags by the money at risk, explains each one, and lets a planner send the fix back to the system that got it wrong.

The demo starts on a fixed **starting day**: six items at plant UBD for sales day Saturday 12 September 2026, with the plant clock at Friday 20:00. Item 76108 reconciles (the control). Five items carry eight designed flags. Every change made in the simulator, and every scenario, starts from there.

The systems it checks

System	What it holds	In this guide
BUSS	Customer sales orders	Sales orders
Oracle	The item master and the demand forecast	Forecast, Item master
ODI	The batch data mapping: each order or forecast linked to a work order and its batches	Batch mapping
WMS	Warehouse stock: on hand, available, allocated to orders	Inventory
MIPS	Production planning: the work order the plant makes	Work orders

The one equation behind every check

Work order = (sales orders + forecast) - allocated stock - available stock

Example, item 76108 (white hot dog buns): (14,339 ordered + 41,769 forecast) - 473 allocated - 7,000 available = **48,635** to make. When every system agrees with this, the item **reconciles** and does not appear in the worklist.

A second layer: machine learning

The equation catches systems that disagree. It cannot catch a wrong value that every system agrees on, such as an order keyed with an extra zero that BUSS, ODI and MIPS all carried through. The **ML layer** compares today with each item's own history of earlier sales days and flags values far outside their normal range (chapter 10).

Two applications

Application	Who uses it	What it is for
Exception worklist	Planners, supervisors, managers	See the flags, understand them, decide, and send corrections. In production this is the only app.
Source simulator	A demo operator	Stands in for BUSS, Oracle, ODI, WMS and MIPS during the proof of concept: change their data, play scenarios, break the integration on purpose, and watch the worklist react.

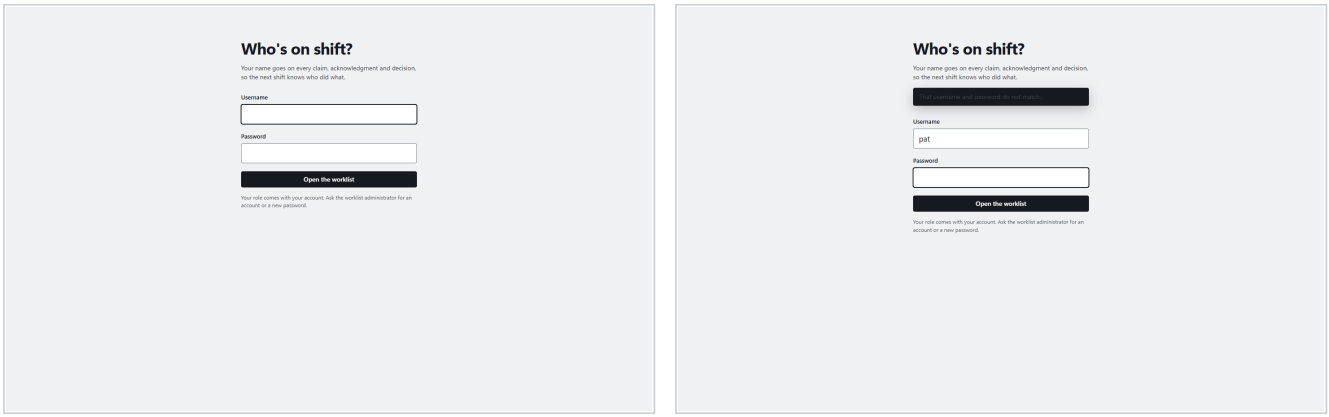
2. Getting started

Roles

Role	Can do
Supervisor	Everything in the worklist except approving corrections: read, claim, acknowledge, note, accept and reject flags, ask the assistant.
Planner	Everything a supervisor can, plus approve and send corrections to the source systems.
Manager	The same as a planner.
Simulator operator	Any account can sign in to the simulator (the same accounts as the worklist).

Sign in

Open the worklist address in a browser. Anyone not signed in goes to the sign-in page. Enter your username and password and click **Sign in**. A wrong password shows 'That username and password do not match'; after repeated failures the sign-in pauses for 15 minutes.



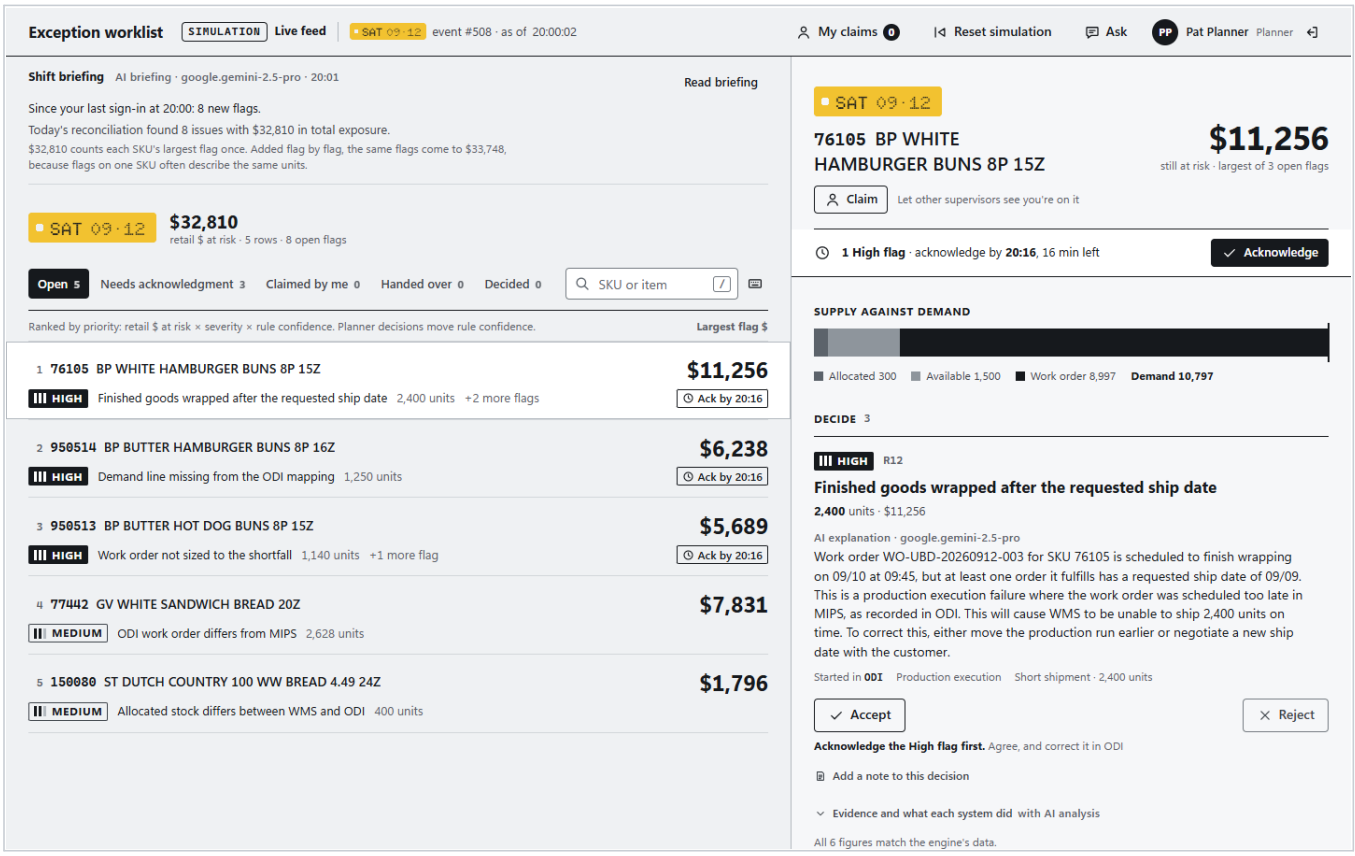
Left: the sign-in page. Right: a wrong password.

The top bar



The worklist's top bar.

Part	Meaning
SIMULATION · Live feed	The data comes from the live (simulated) event feed. In workbook mode it says 'Source workbook, not live' instead.
SAT 09·12	The sales day being reconciled, as a coloured day tag (each weekday has its colour).
event #74 · as of 20:00:41	The last change received, and the plant time the screen last checked.
Rows changed · Update	Appears when the data or other people changed something. See section 3.8.
My claims	How many SKUs you have claimed; click to list them.
Reset simulation	Back to the starting day for everyone (simulation only).
Ask	Opens the assistant.
Your initials, name and role	Who is signed in. The arrow signs you out.

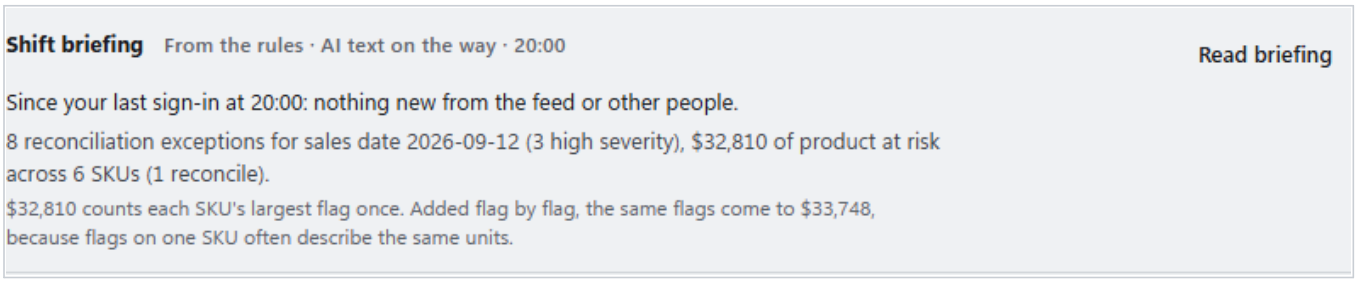


The worklist on the starting day: briefing, day tile, view tabs and the ranked queue on the left; the open SKU on the right.

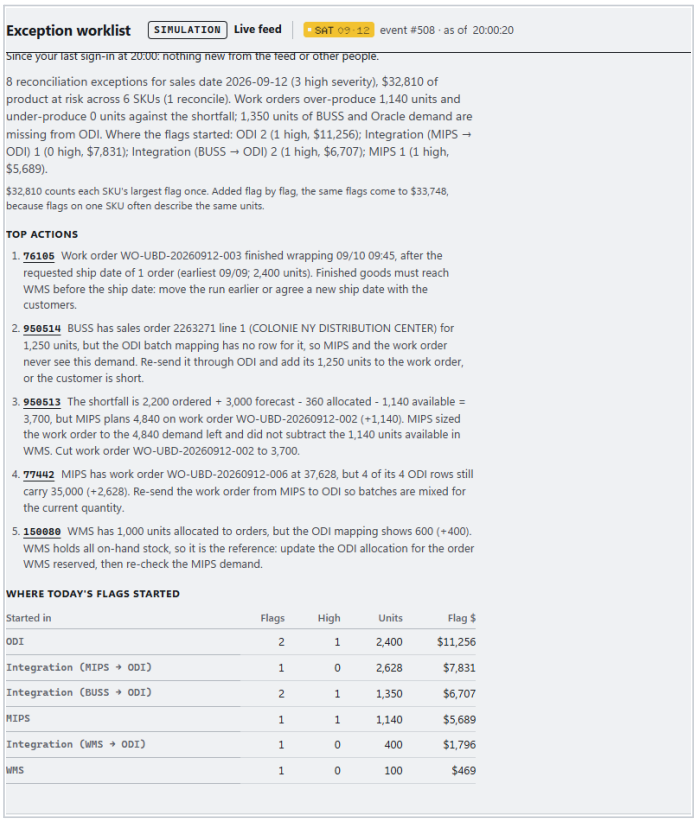
3. The exception worklist

3.1 Shift briefing

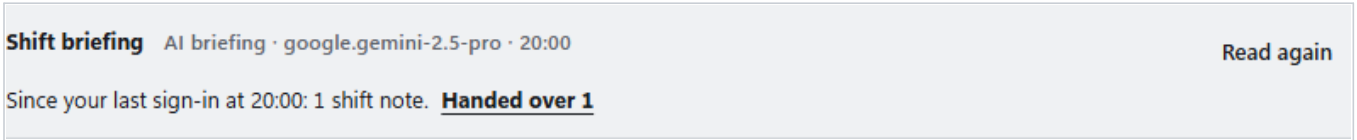
The briefing at the top sums up the day in one line: how many exceptions, how many High, and the money at risk. It is written by the rules first and replaced by an **AI briefing** about 20 seconds later (the label says which). Click **Read briefing** for the full text and the top actions, one line per SKU; **Mark read** folds it away until something changes.



The briefing, written by the rules while the AI text is on its way.



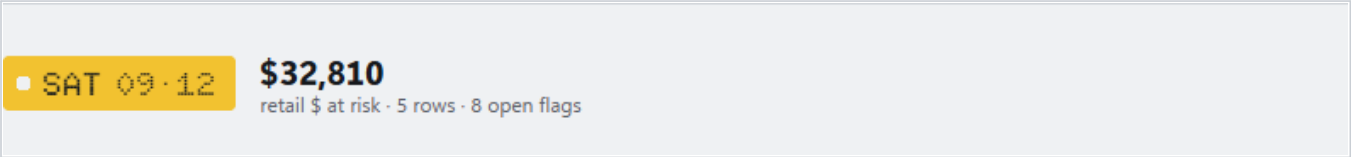
Read briefing: the full text and the top actions.



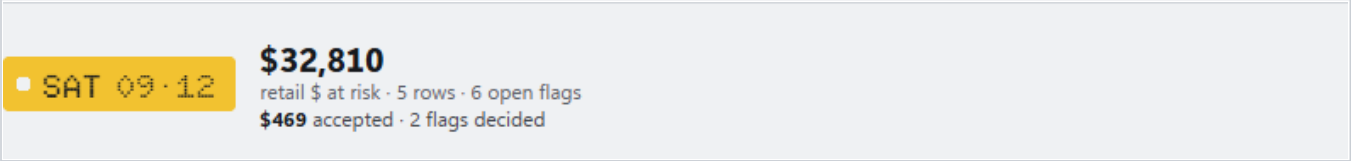
After you sign in again, the briefing says what others did since your previous sign-in, and links to Handed over.

3.2 The sales-day tile

The tile shows the money still at risk for the day (each SKU counted once, by its largest open flag), how many rows and open flags remain, and, after decisions, what was accepted and decided.



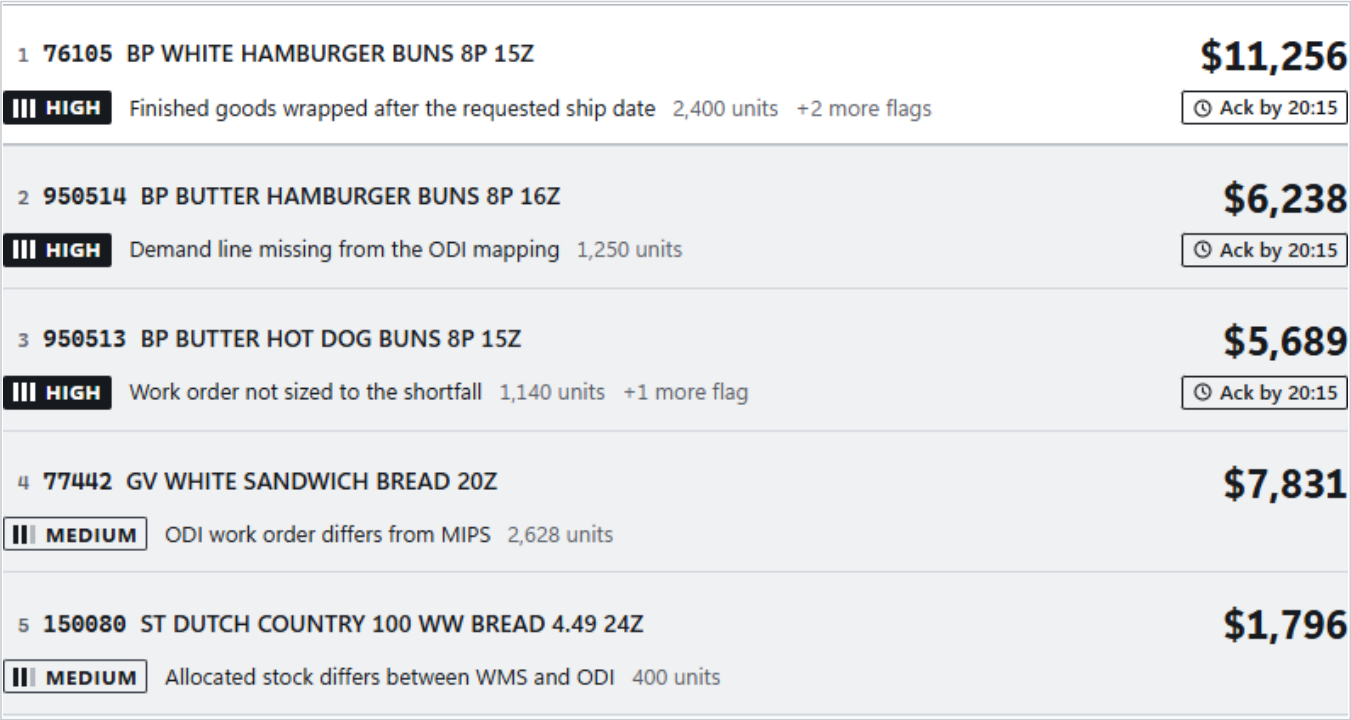
The starting day: $32,810 at risk across 5 rows and 8 open flags.



After two decisions: accepted dollars and decided flags are added.

3.3 The queue

Each row is one SKU with open flags, ranked by **priority**: retail dollars at risk × severity × rule confidence. The row shows its rank, item, largest flag, severity, units, how many more flags it has, the acknowledgment deadline for High flags, and the dollars.



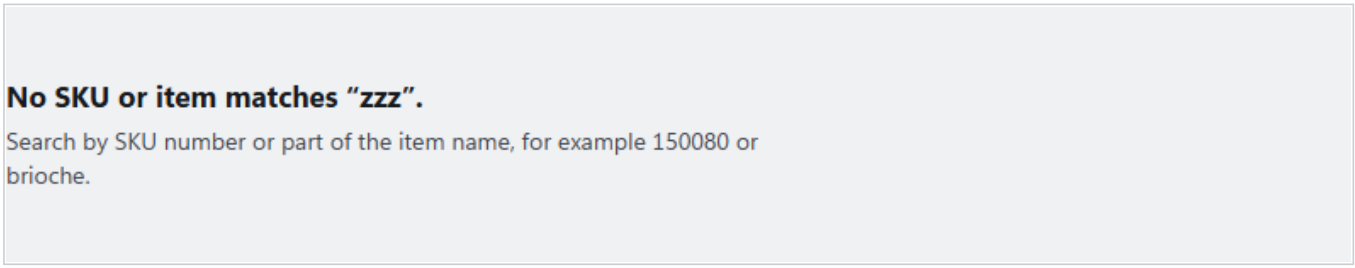
Five rows: 76105 ($11,256, High), 950514 ($6,238, High), 950513 ($5,689, High), 77442 ($7,831, Medium) and 150080 ($1,796, Medium).

3.4 View tabs and search

Tab	Shows
Open	Every SKU with open flags (the default).
Needs acknowledgment	SKUs with a High flag not yet acknowledged.
Claimed by me	SKUs you have claimed.

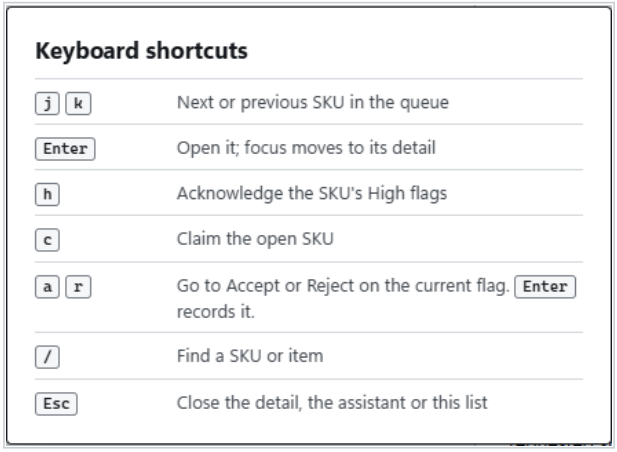
Tab	Shows
Handed over	Open work someone already touched: a shift note, someone else's claim, or a High flag acknowledged but not decided.
Decided	Accepted and rejected flags (the count is flags, not rows).

Type in **SKU or item** to search by SKU number or part of the name (for example 950513 or hamburger). An empty view or search always says why it is empty; it never says everything reconciles while flags are open.



A search with no match.

3.5 Keyboard shortcuts

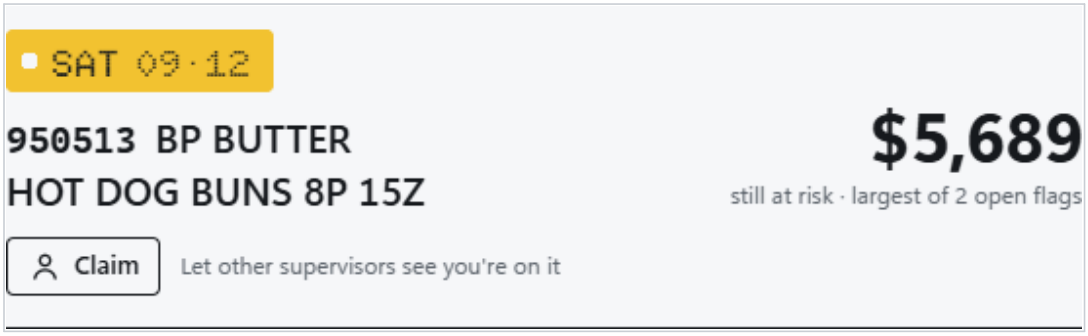


Press **?** (or click the keyboard icon) to see the shortcuts. They work when you are not typing in a box.

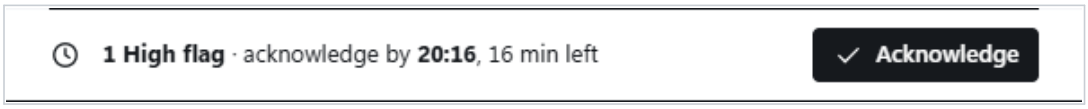
- **j** / **k**: next or previous SKU
- **Enter**: open it
- **h**: acknowledge the SKU's High flags
- **c**: claim the open SKU
- **a** / **r**: go to Accept or Reject; Enter records it
- **/**: search
- **Esc**: close the detail, the assistant or the list

3.6 The SKU detail

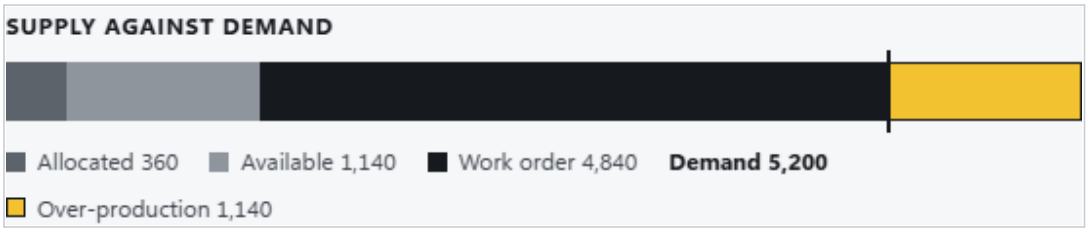
Click a row (or press Enter) to open its detail on the right. From top to bottom:



Header: the day tag, SKU and item, the money still at risk, and Claim.

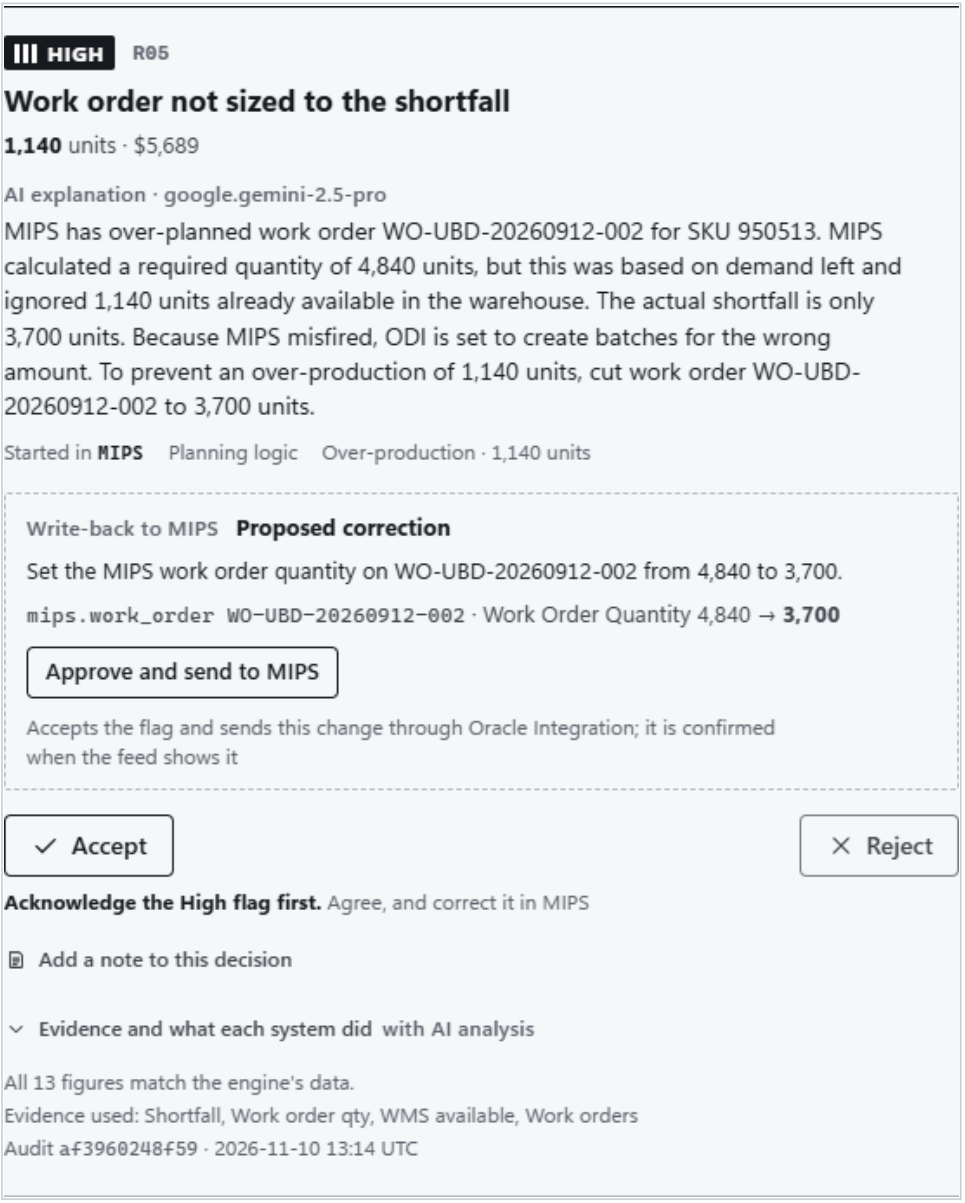


Acknowledgment bar: a High flag must be acknowledged within 15 minutes of being raised.



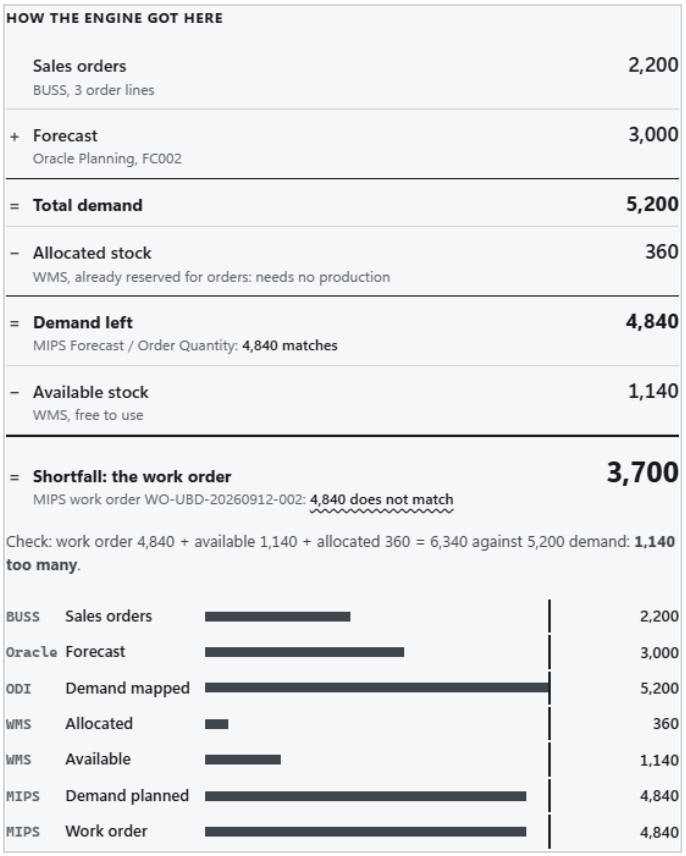
Supply against demand: allocated, available and the work order against the demand mark. Over-production shows past the mark in yellow; a shortfall shows hatched.

Decide lists the SKU's flags, most important first. Each flag shows its severity and rule, a plain-language title, the units and dollars, the explanation, where it started, any proposed correction, and Accept / Reject.



A flag: R05, the work order was not sized to the shortfall.

How the engine got here works through the equation for the SKU with each system's figure, and marks where a system does not match. Below it, each system's quantity is drawn on the same scale.





Source records (at the bottom) lists the BUSS orders, the forecast, the ODI rows, the WMS stock and the MIPS work order behind the figures, as each system holds them.

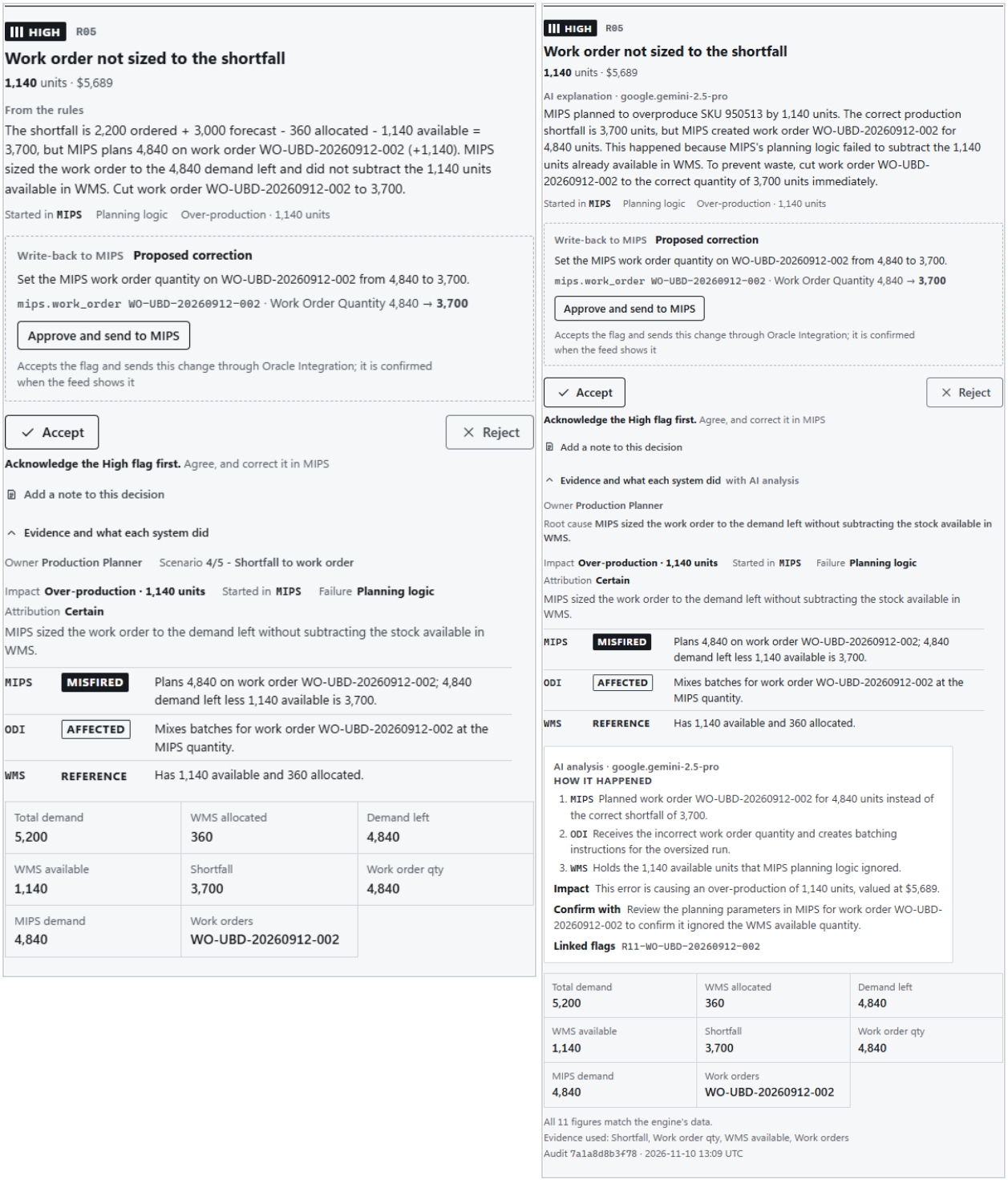


Source records for 950513.

3.7 Evidence, accountability and AI explanations

Open **Evidence and what each system did** under a flag to see who owns the problem, the root cause, where it started, the failure type, and what each system did: **Misfired** (made the mistake), **Affected** (received it), **Reference** (holds the right figure) or **To check**. The raw evidence values follow.

When the AI text arrives, each flag's explanation is replaced by an **AI explanation**, labelled with the model, and the evidence adds the AI's step-by-step account. Every figure the AI quotes is checked against the engine ('All 13 figures match the engine's data') and the call is kept in an audit log (the audit id is shown).



Left: evidence and accountability, written by the rules. Right: the same flag with the AI explanation, its analysis, the figure check and the audit id.

3.8 Staying current: the Update pill

When a source system changes, or another person claims, notes or decides, the top bar shows **N rows changed by others · Update** within a second. Nothing on your screen moves until you click it, so a row never jumps away while you read. **Update** redraws the queue, the day tile, the briefing and the open detail. If you are typing a note, the detail waits until you send or clear it.



The Update pill in the top bar.

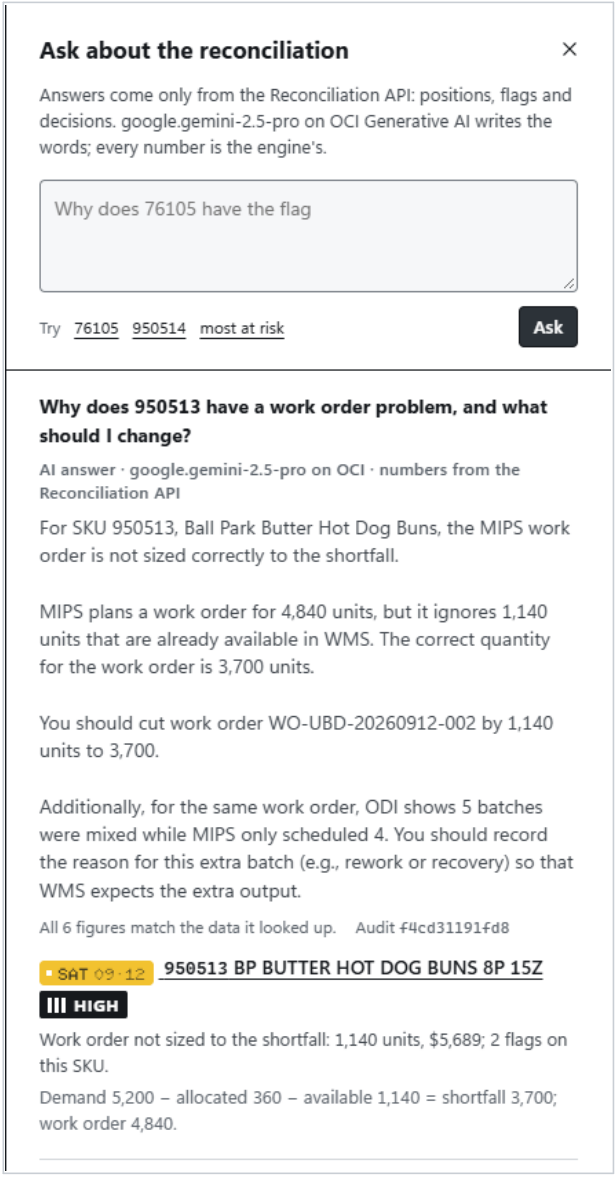
- **AI explanations ready · Update**: the AI text arrived.
- **Figures changed · Update**: totals moved without a row changing.
- **Simulation reset to the starting day · Reload**: someone reset the simulation from the worklist.
- Your own actions never show the pill.

> **Before you decide, update**
>
> If the black pill is showing, click Update first, so you decide on the current numbers.

3.9 Ask the assistant

Click **Ask** in the top bar, type a question, and send it. The assistant answers from the current data (the figures come from the Reconciliation API), names the systems involved and says what to change. Suggested questions about the top of the queue are offered under the box.

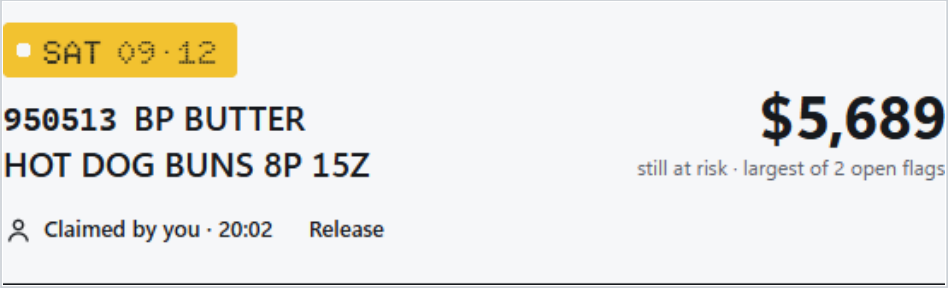
Example tested: 'Why does 950513 have a work order problem, and what should I change?' The answer gave the 3,700 shortfall, the 1,140 over-production and the cut to make in MIPS.



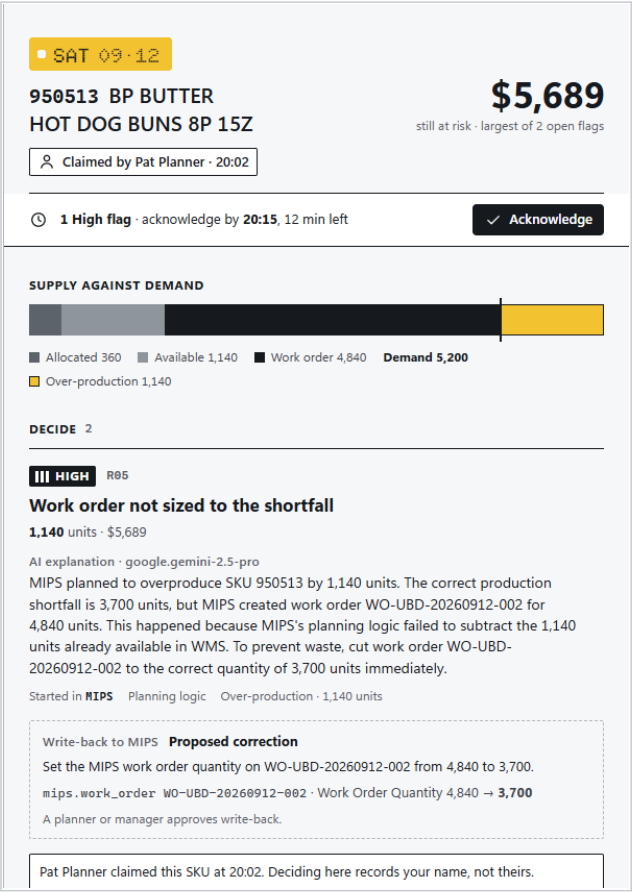
4. Working as a team

4.1 Claim a SKU

Click **Claim** on a SKU so other people can see you are working on it. Your **My claims** count goes up, and the SKU appears under **Claimed by me**. Others see your initials on the row and, in the detail, a warning that deciding records their name, not yours. Click **Release** to give it back.



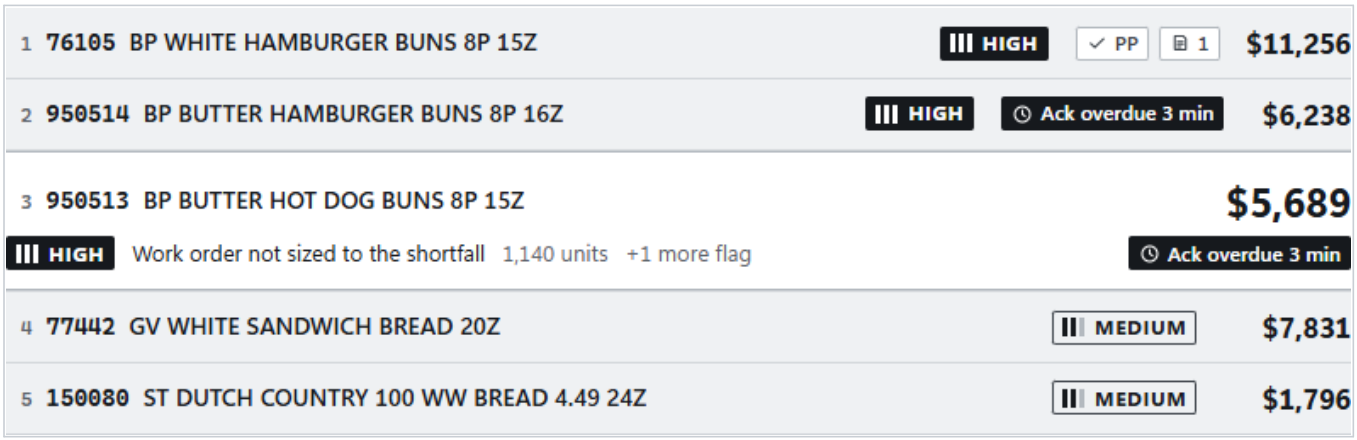
Pat has claimed 950513.



What Sam sees on 950513 after clicking Update: Pat's claim and the warning.

4.2 Acknowledge High flags

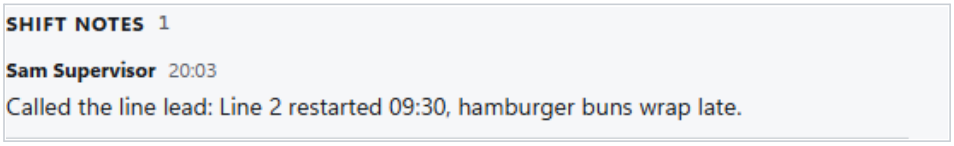
A High flag must be acknowledged within 15 minutes of being raised (plant time). Click **Acknowledge** on the bar at the top of the detail (or press h). While a High flag waits, every Accept on that SKU steps back to an outline, so acknowledging is the first thing to do. Unacknowledged flags turn **overdue**.



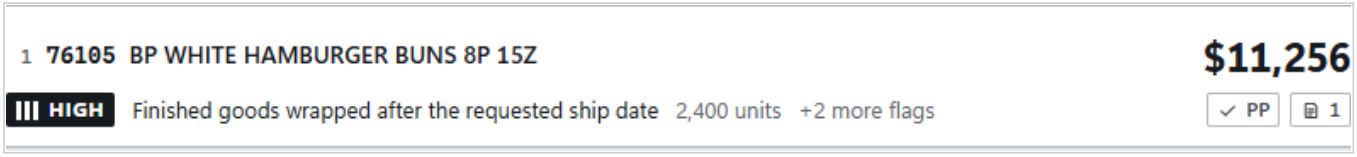
76105 acknowledged by Pat (tick and initials, 1 shift note); 950513 is overdue.

4.3 Shift notes and Handed over

Open **Leave a note for the next shift** at the bottom of the flags, write what you checked or are waiting on, and send it. Notes show on the SKU with the author and time. The **Handed over** tab collects open work that someone has touched but not finished.



Sam's note on 76105.



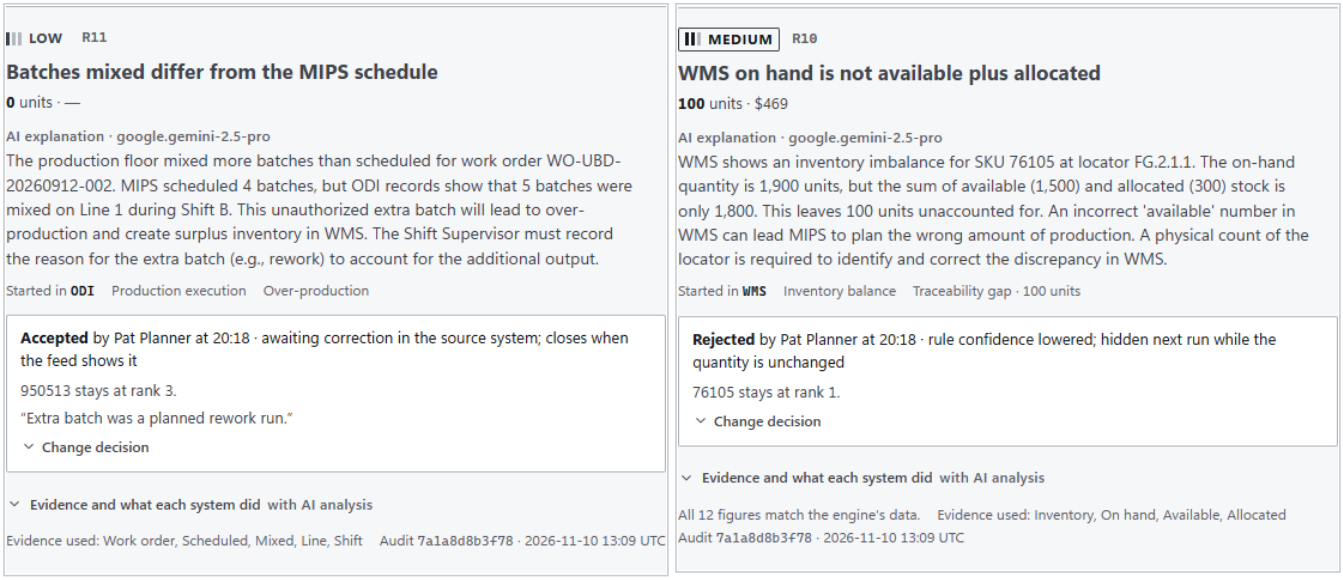
Pat's Handed over tab after Sam's note.

5. Deciding on a flag

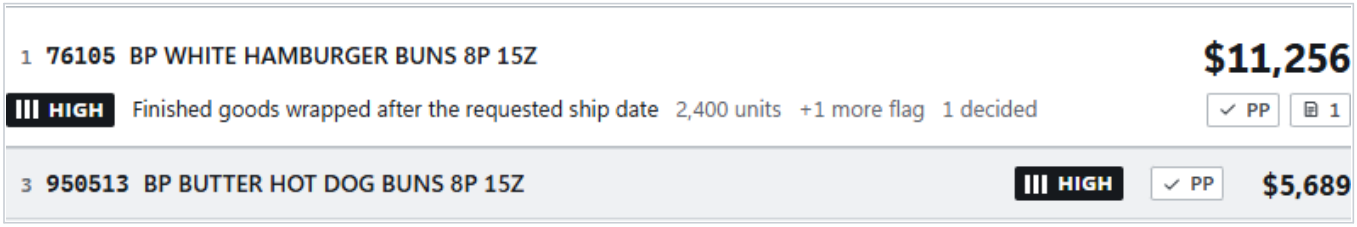
Every flag needs a decision. **Accept** means the flag is right and the source system must be corrected; **Reject** means the flag is wrong. You can add a note to a decision ('Add a note to this decision').

Decision	What happens
Accept	The flag shows 'Accepted by you, at time', awaiting correction in the source system. It closes by itself when the feed shows the systems agree. The rule's confidence goes up, so its flags on other rows rank higher.
Reject	The rule's confidence goes down a little and the flag stays hidden on later runs while its quantity is unchanged.
Change	Open 'Change' on a decided flag to switch the decision.

After a decision the detail says how the row moved in the queue and why (it now ranks on its next flag; the rule's new confidence re-ranks other rows).



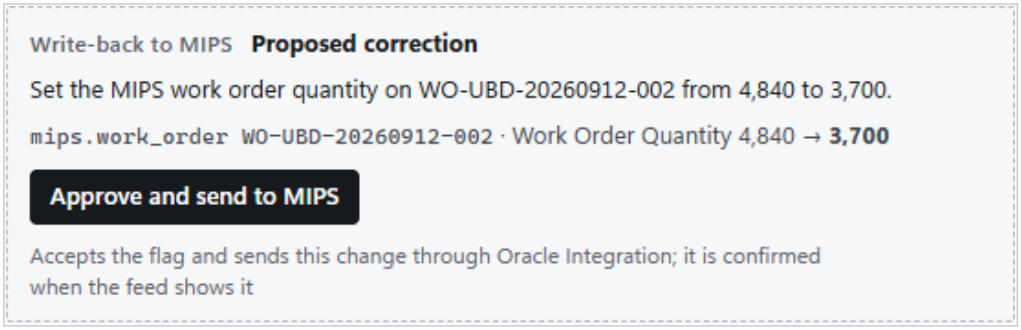
Left: R11 accepted with a note. Right: R10 rejected.



The Decided tab: 2 flags decided.

6. Sending corrections to the source systems

For most rules the agent proposes the exact change that fixes the source system, shown in a dashed box under the flag: which system, which row, which column, from what to what.

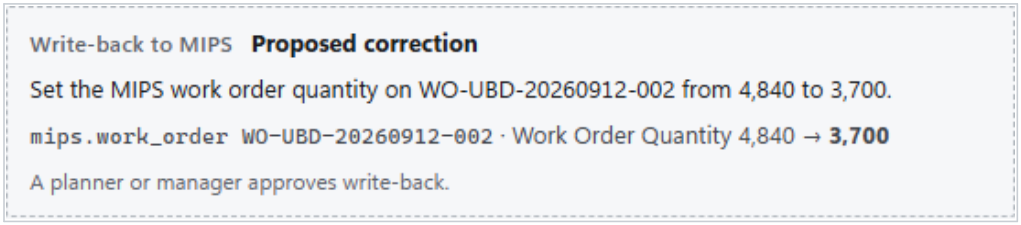


A proposed correction: MIPS work order 4,840 -> 3,700.

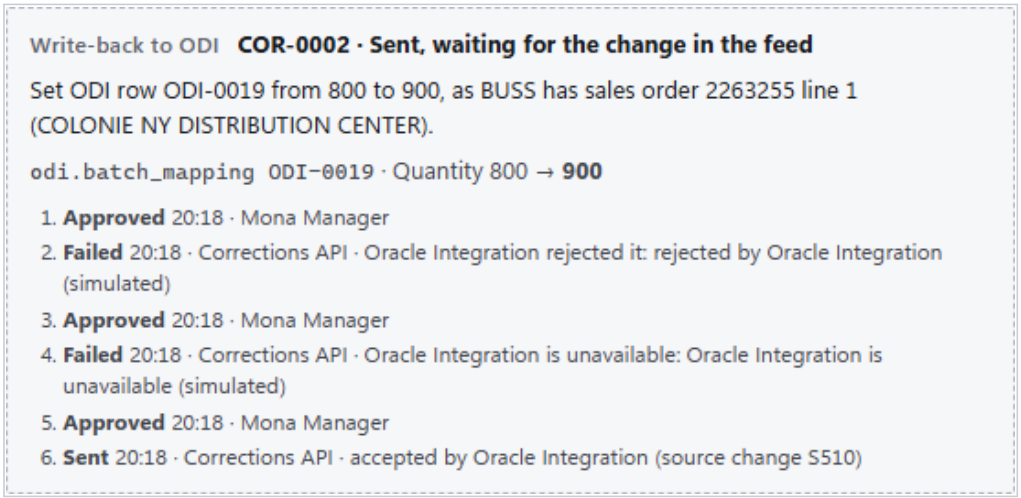
6.1 Approve and send

1	A **planner or manager** clicks **Approve and send to <system>**. This also accepts the flag.
2	The correction goes to Oracle Integration (in the demo, the simulator's corrections endpoint). Its status is **Approved**, then **Sent, waiting for the change in the feed**.
3	The source system applies the change, and the change comes back through the feed like any other.
4	The correction becomes **Confirmed in the feed** and the flag closes, because the systems now agree.

Supervisors see the proposal with 'A planner or manager approves write-back.' and no button.

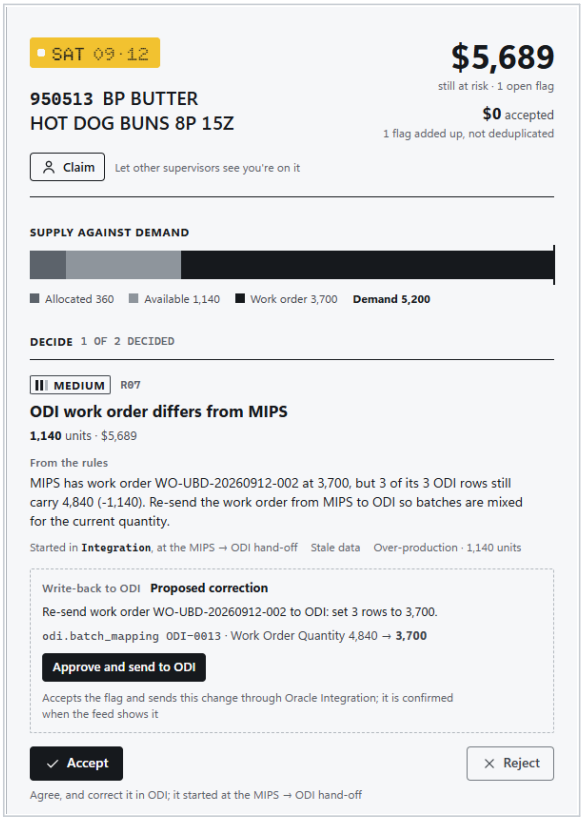


What a supervisor sees.



Sent and waiting: the integration is holding the change (here a 120 s delay).

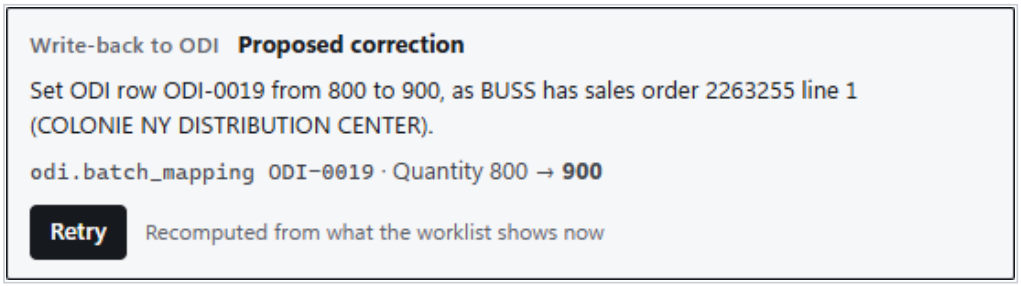




Left: the simulator's MIPS work orders after the correction (3,700). Right: 950513 afterwards: R05 is gone and there is no over-production.

6.2 When a correction fails

If Oracle Integration refuses the change or is unavailable, the correction fails and a **Retry** button appears. Retry recomputes the change from what the worklist shows now and sends it again.



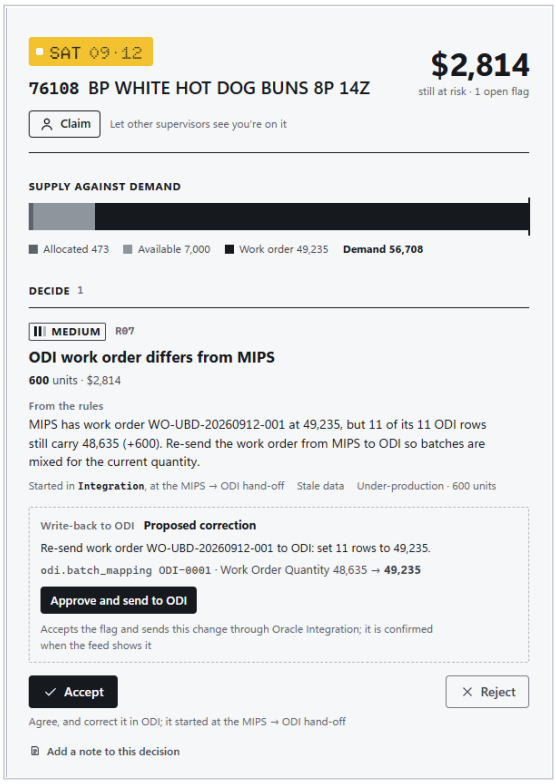
A failed correction with Retry.

Known issue: while a correction is failed, its reason is not shown

The worklist records each failure and its reason (for example 'Oracle Integration rejected it' or 'Oracle Integration is unavailable'). But while the correction is in the Failed state, the SKU detail shows it as a plain 'Proposed correction' with Retry, without the word Failed or the reason. The reasons appear in the correction's history once it is retried and sent (see the screenshot in 6.1), and at any time through the API (GET /api/corrections). See the test report, checks E4 and E5.

6.3 A full chain: fixing every system for one item

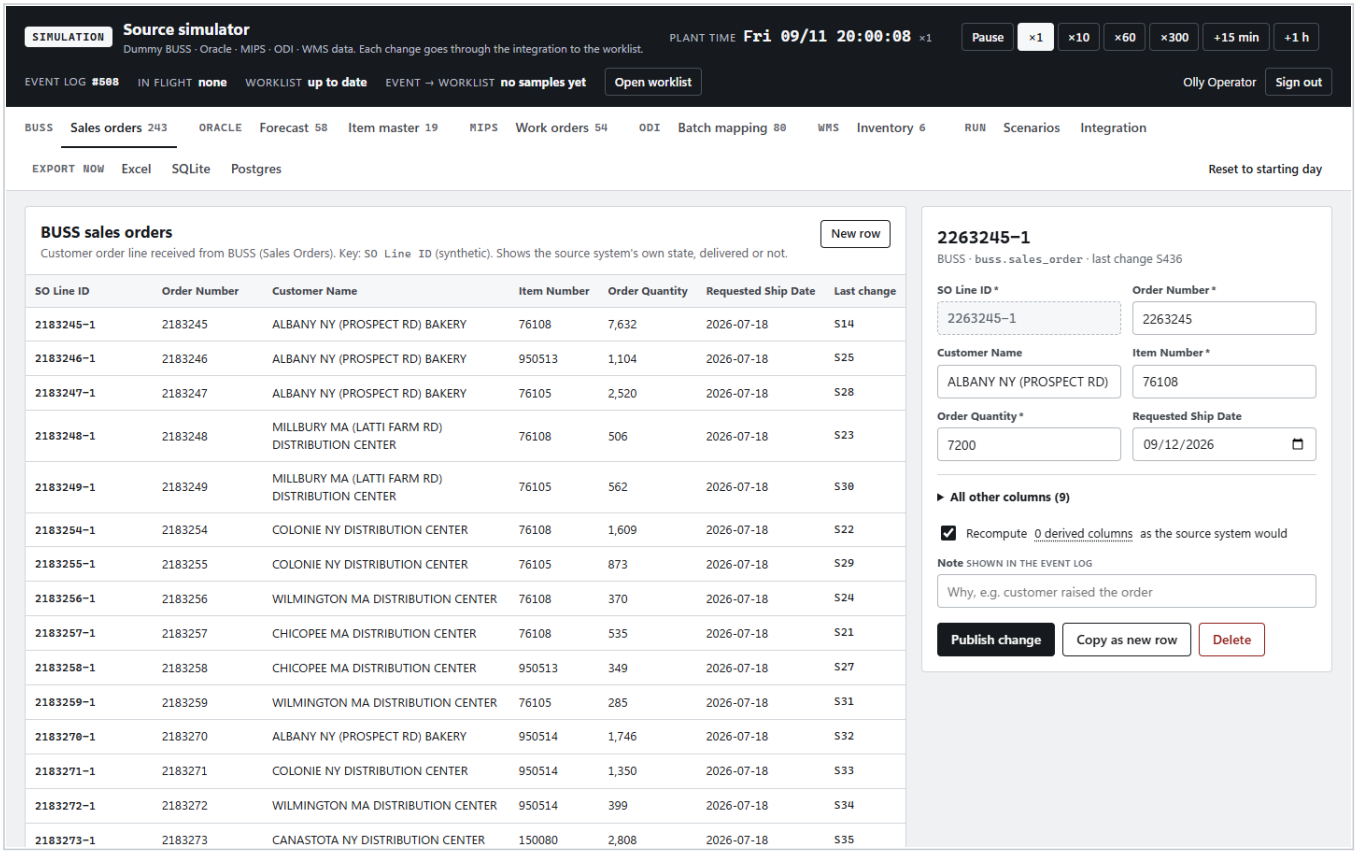
A problem can cross several systems. After Scenario 1 (a customer raised an order that ODI never received), 76108 had three flags. Approving them in turn fixed each system: R02 (ODI), then R04 (MIPS demand), then R05 (MIPS work order). Each next proposal was recomputed from the corrected data, and after the third, 76108 reconciled and left the queue.



76108 after the three corrections.

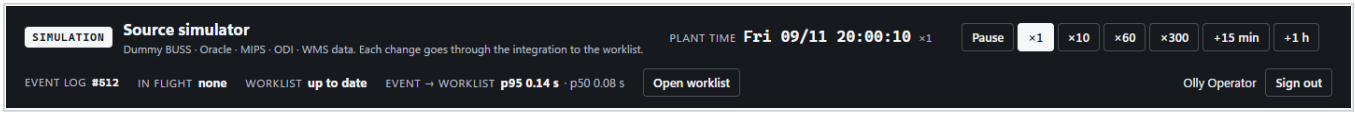
7. The source simulator

The simulator stands in for the five source systems during the proof of concept. Sign in with the same account as the worklist. Keep the worklist open beside it to watch every change arrive.



Tables on the left, the row editor on the right.

7.1 The status bar



Part	Meaning
PLANT TIME	The simulated plant clock and its controls (section 7.5).
EVENT LOG #	How many changes have been published.
IN FLIGHT	Changes the integration is still holding (delayed or out of order).
WORKLIST	Whether the worklist has processed everything ('up to date') or how far behind it is.
EVENT → WORKLIST	How long changes take to reach the worklist (p50 and p95; 0.03 s / 0.09 s in the test).
Open worklist	Opens the worklist in a new tab.

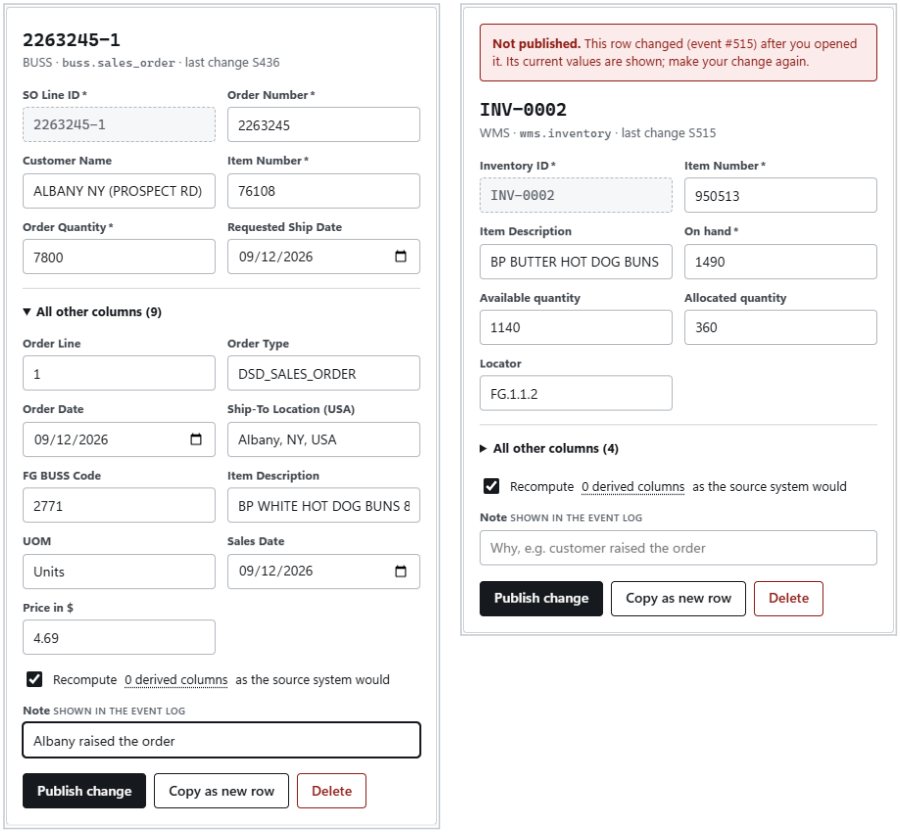
7.2 Tables

The tabs list each system's tables with their row counts: BUSS Sales orders; Oracle Forecast and Item master; MIPS Work orders; ODI Batch mapping; WMS Inventory. Tables show each system's own data, whether or not it

has reached the worklist yet. The **Last change** column gives the change number of each row.

7.3 Change, add and delete rows

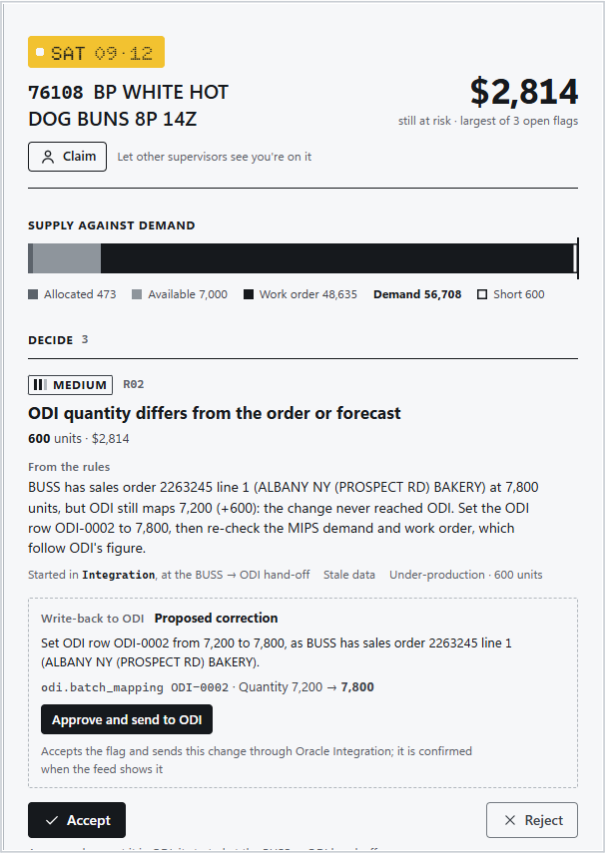
1	Click a row's key to open it in the editor. The main columns are shown; 'All other columns' opens the rest.
2	Change the values, optionally write a **Note** (shown in the event log), and click **Publish change**.
3	**Copy as new row** opens a new row prefilled from this one: give it a new key and click **Insert row**. **New row** (above the table) starts from empty.
4	**Delete** removes the row after a confirmation; a delete event is published.
5	If someone else changed the row after you opened it, your change is refused: 'This row changed (event #N) after you opened it. Its current values are shown; make your change again.'



Left: changing a BUSS order to 7,800. Right: an older form refused.

7.4 What the worklist does with each change (tested)

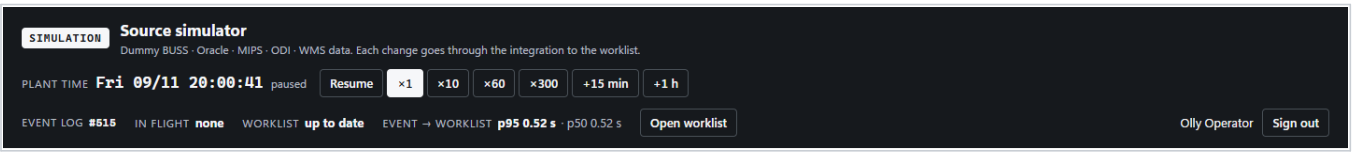
Change in the simulator	What the worklist showed
BUSS 2263245-1 quantity 7,200 -> 7,800	'1 row changed by others · Update'; 76108 (which reconciled) appeared with R02, R04 and R05, 600 units each.
WMS INV-0001 on hand 7,473 -> 7,373	R10-INV-0001: on hand is not available + allocated, 100 units.
Oracle forecast FC001 41,769 -> 42,769	R02-FC001 (ODI still maps the old forecast) and larger R04/R05.
New BUSS order 2263299-1 (Copy as new row), 500 units	R01-2263299-1: the order is missing from ODI.
Delete 2263299-1	R01-2263299-1 cleared.
ODI-0013 batches mixed 5 -> 6	No flag moved, but the pill still showed; R11 now said the ODI rows disagree (ODI-0013: 6 of 4).



76108 in the worklist after the BUSS change.

7.5 The plant clock

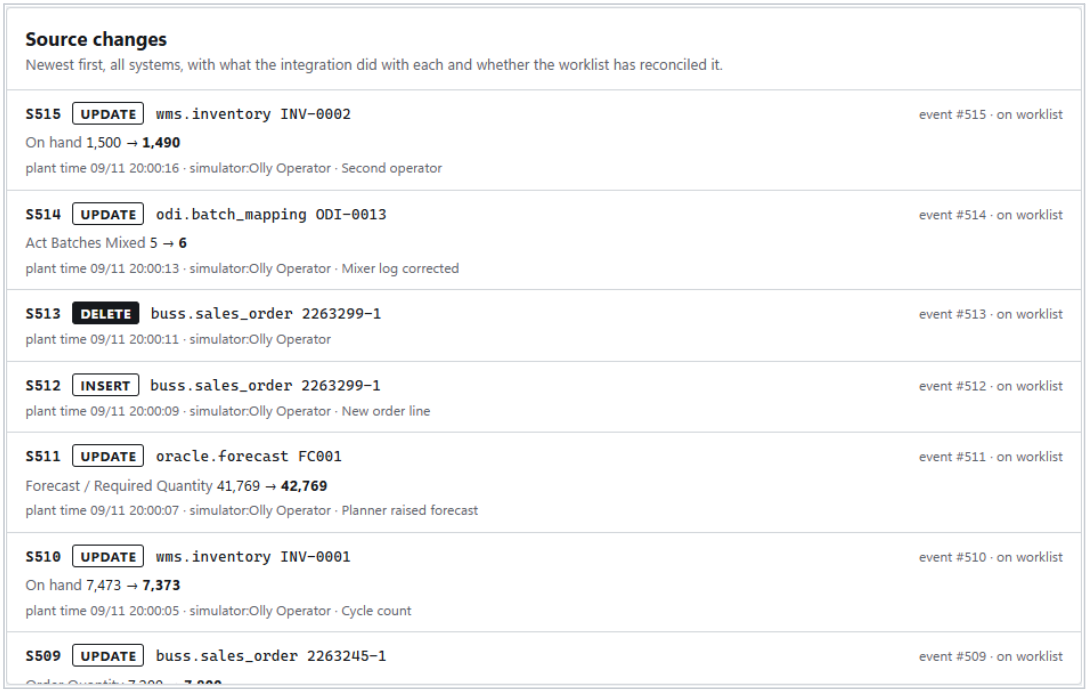
Pause / **Resume** stop and restart plant time; **×1, ×10, ×60, ×300** set its speed; **+15 min** and **+1 h** jump ahead. The worklist's as-of time and the 15-minute acknowledgment deadlines follow plant time, so +15 min is a quick way to see a High flag go overdue.



The clock paused.

7.6 Source changes, exports and reset

- **Source changes** (under every table) lists each change, newest first: operation, row, what changed, plant time, who made it, the note, and whether the worklist has reconciled it.
- **Export now**: **Excel** (every source table, the reconciled position and the flags), **SQLite** (with the v_sku_position view, for tools such as DBeaver) or **Postgres** (a script that loads schema recon). Each is built from the current rows.
- **Reset to starting day** clears every change, fault and delivery and publishes the starting day again.



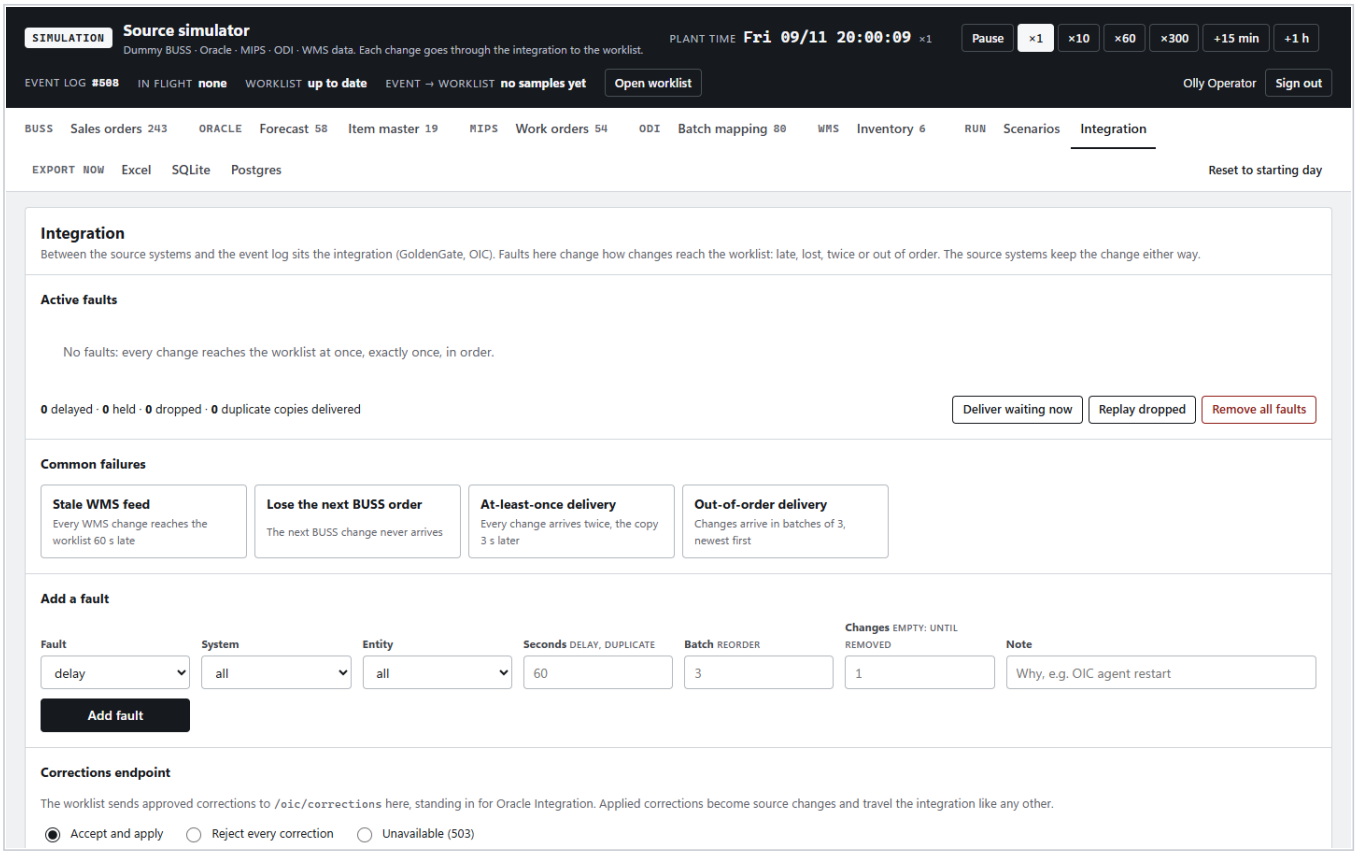
Source changes.

Known issue: use the worklist's Reset simulation

The simulator's 'Reset to starting day' resets the data but not the worklist's decisions: flags decided before the reset come back as 'Accepted - awaiting correction' and never close. Use **Reset simulation** in the worklist's top bar instead; it clears data, decisions, claims, acknowledgments and notes for everyone and tells open screens. See the test report, check K1.

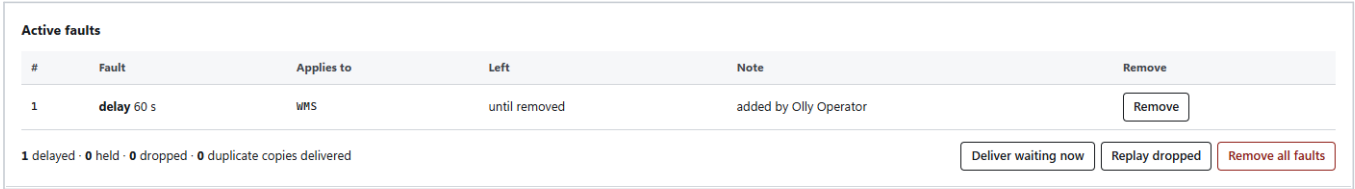
8. Integration faults

Between the source systems and the worklist sits the integration (GoldenGate and Oracle Integration in production). The **Integration** tab breaks it on purpose so you can see how the worklist copes. The source systems keep every change either way.



The Integration tab.

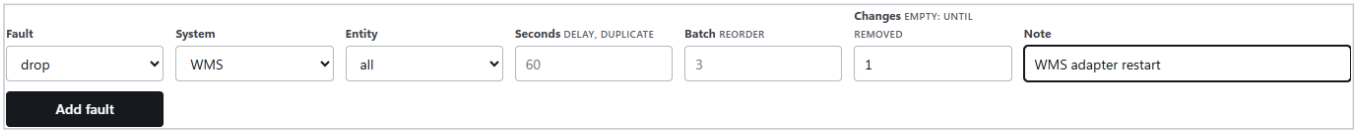
Fault	What it does	Tested result
Stale WMS feed (delay)	Every WMS change reaches the worklist 60 s late.	The worklist stayed quiet while the change waited (1 delayed); 'Deliver waiting now' released it and R10-INV-0002 appeared.
Lose the next BUSS order (drop)	The next BUSS change never arrives.	The worklist never saw it (1 dropped) until 'Replay dropped' sent it again from the source.
At-least-once delivery (duplicate)	Every change arrives twice, the copy 3 s later.	The worklist applied it once (R10-INV-0002 at 50 units, not doubled).
Out-of-order delivery (reorder)	Changes arrive in batches of 3, newest first.	Three edits arrived newest first; the worklist still ended on the last value (950513 orders 2,330).



A change held by the delay fault.

Custom faults and the buttons

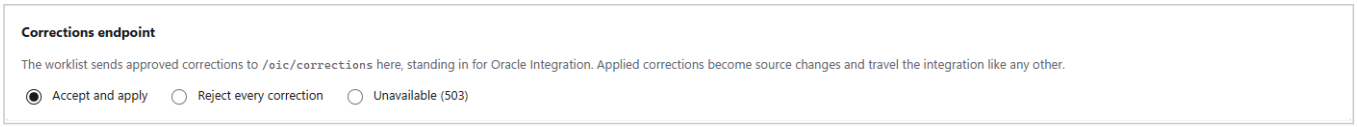
- **Add a fault**: choose the kind (delay, drop, duplicate, reorder), the system and entity, the seconds, batch size, how many changes it applies to (empty: until removed) and a note.
- **Deliver waiting now** releases delayed and held changes. **Replay dropped** resends dropped changes from the source, in order. **Remove all faults** ends every fault; **Remove** ends one.



Adding a custom fault.

The corrections endpoint

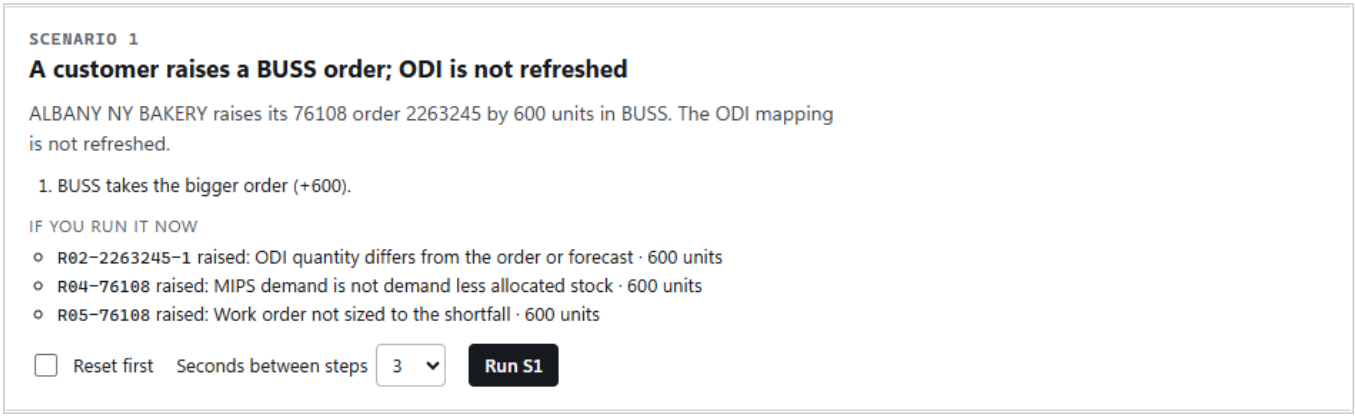
This is where the worklist sends approved corrections. **Accept and apply** (normal), **Reject every correction**, or **Unavailable (503)** let you test the failure paths in section 6.2.



9. The rule scenarios (S1-S10)

Each scenario is a short script that breaks one step of the demand-to-delivery flow for item 76108. Open **Scenarios** in the simulator. Each card tells the story, lists the steps, and shows **If you run it now**: the flags it will raise, worked out on a copy of the current data.

1	Tick **Reset first** to start from the starting day (recommended).
2	Choose the **seconds between steps** (0, 3 or 10).
3	Click **Run S1** (and so on). The Run panel shows each step and its event as it lands, then what it raised.
4	In the worklist, click the **Update** pill and open 76108.



A scenario card.

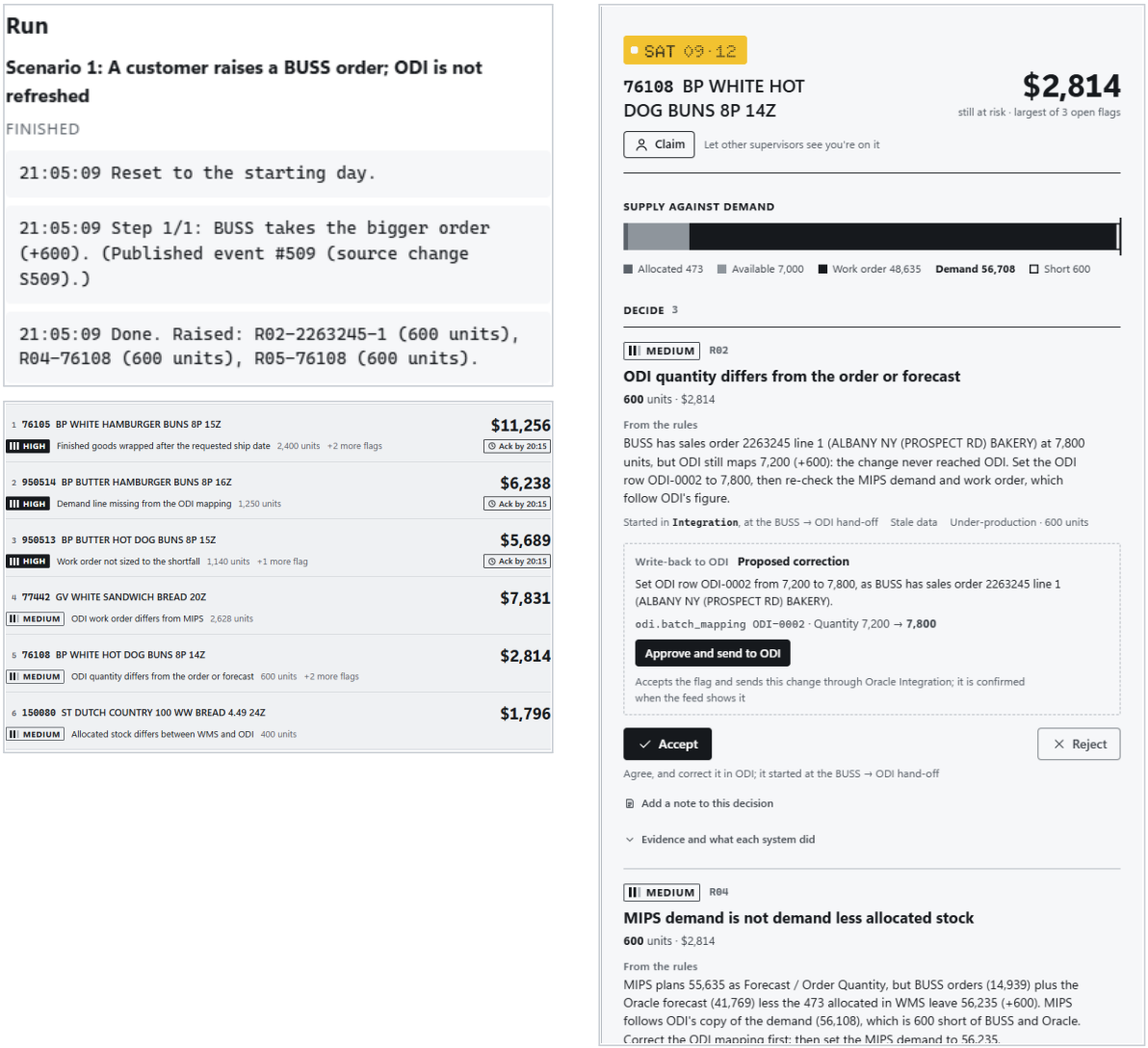
The simulator also has four ML scenarios (S11-S14), covered in chapter 10. All ten rule scenarios were run from the starting day and each raised exactly its expected flags. The pages that follow show the run log, the worklist's queue and the 76108 detail for each.

Scenario 1: A customer raises a BUSS order; ODI is not refreshed

Story. ALBANY NY BAKERY raises its 76108 order 2263245 by 600 units in BUSS; the ODI mapping is not refreshed.

Expected flags. R02 (ODI quantity differs), R04 (MIPS demand), R05 (work order), 600 units each.

Result: PASS. R02-2263245-1 (600 u $2,814), R04-76108 (600 u $2,814), R05-76108 (600 u $2,814)



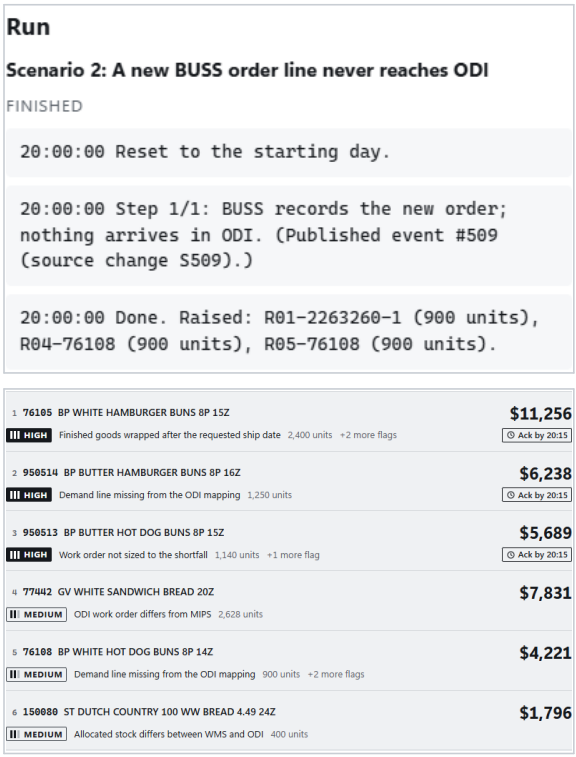
Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

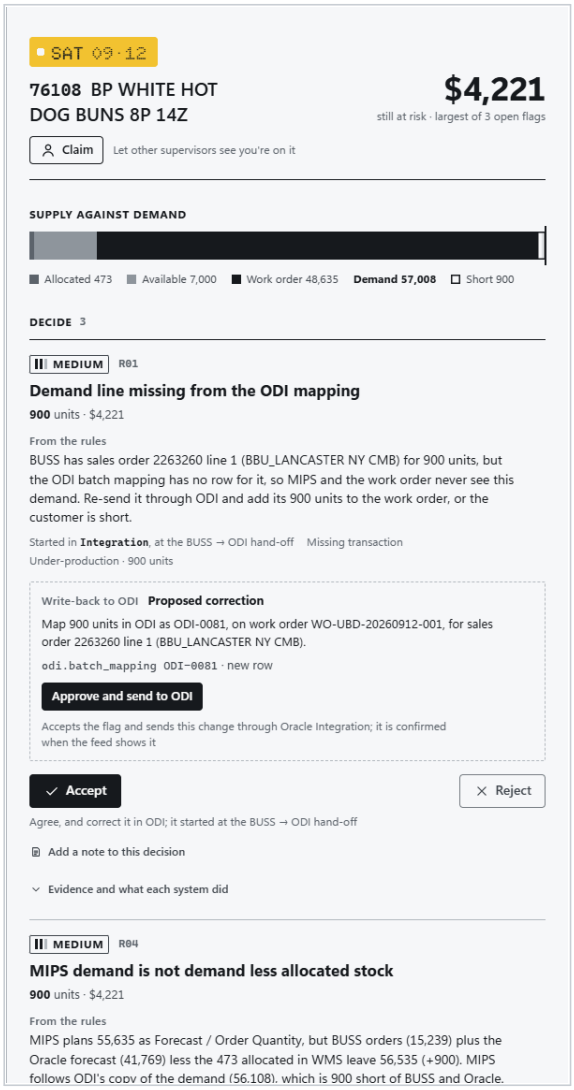
Scenario 2: A new BUSS order line never reaches ODI

Story. BBU_LANCASTER NY CMB places a new 900-unit 76108 order in BUSS; the BUSS-to-ODI interface drops it.

Expected flags. R01 (order missing from ODI), R04, R05: 900 units.

Result: PASS. R01-2263260-1 (900 u $4,221), R04-76108 (900 u $4,221), R05-76108 (900 u $4,221)





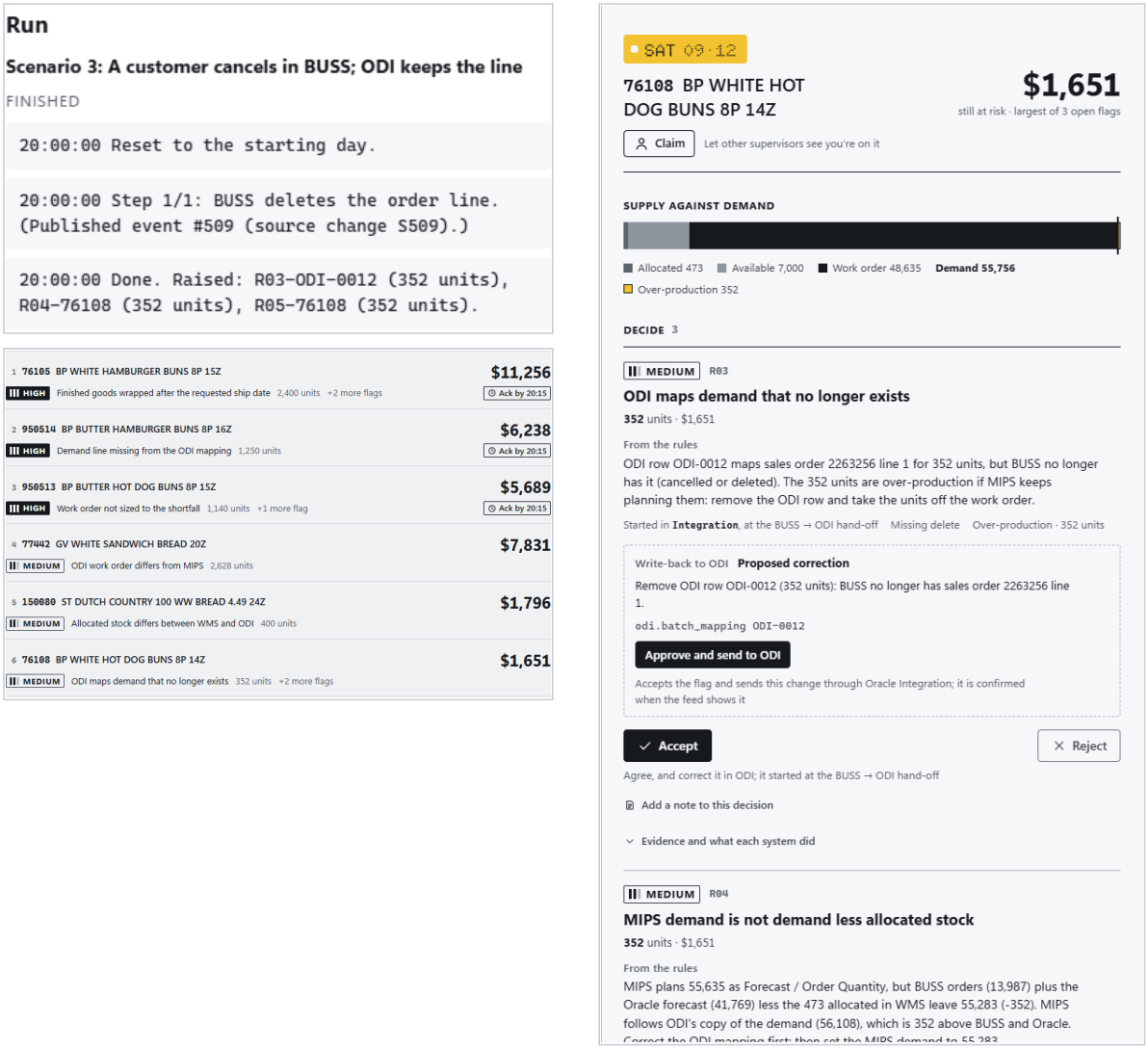
Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

Scenario 3: A customer cancels in BUSS; ODI keeps the line

Story. WILMINGTON MA cancels its 76108 order 2263256 in BUSS; the ODI mapping keeps the line.

Expected flags. R03 (ODI maps an order that no longer exists), R04, R05: 352 units.

Result: PASS. R03-ODI-0012 (352 u $1,651), R04-76108 (352 u $1,651), R05-76108 (352 u $1,651)



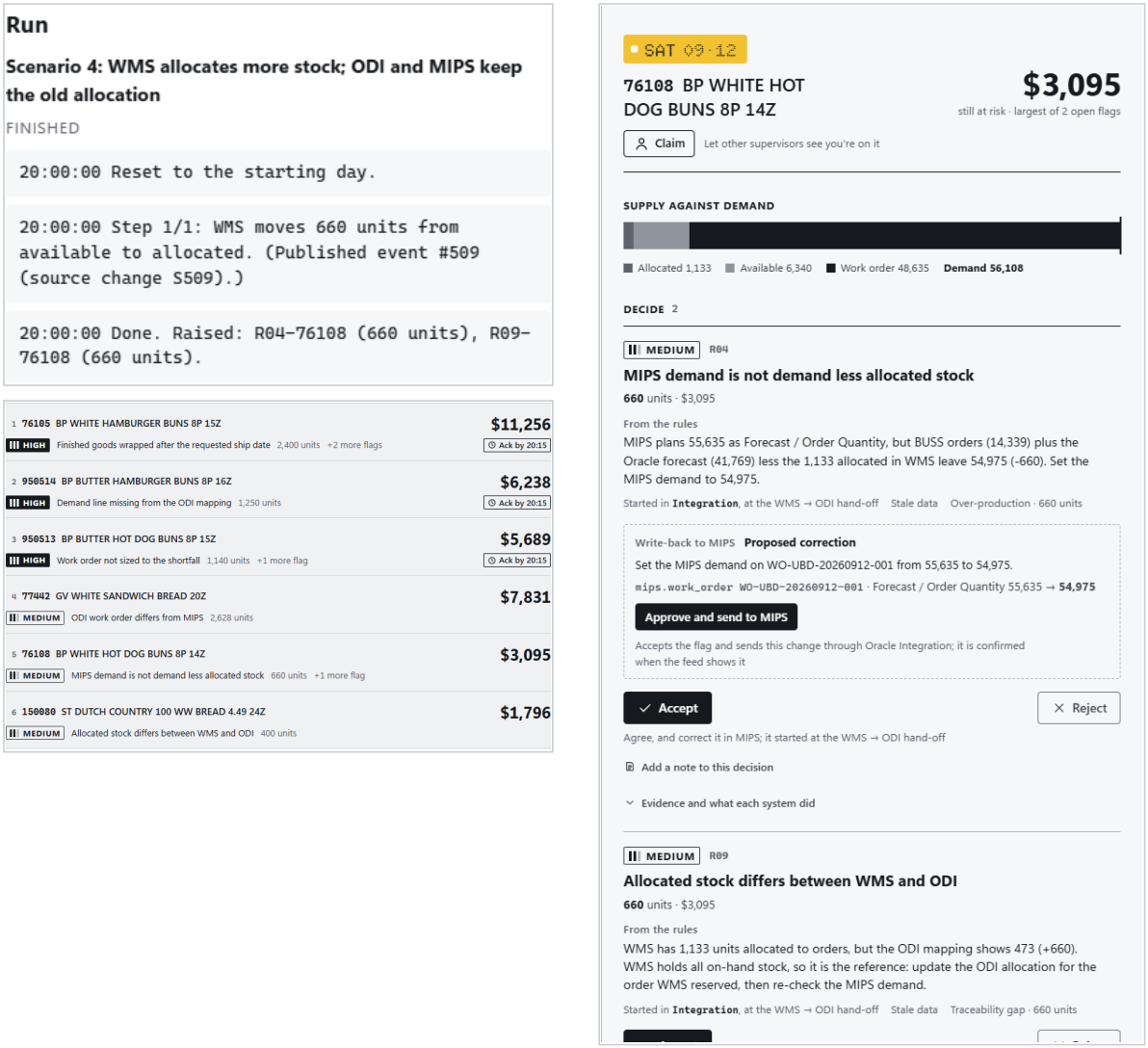
Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

Scenario 4: WMS allocates more stock; ODI and MIPS keep the old allocation

Story. WMS reserves 660 more units of 76108 for an order; ODI and MIPS keep the old allocation.

Expected flags. R09 (allocation differs between WMS and ODI), R04: 660 units.

Result: PASS. R04-76108 (660 u $3,095), R09-76108 (660 u $3,095)



Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

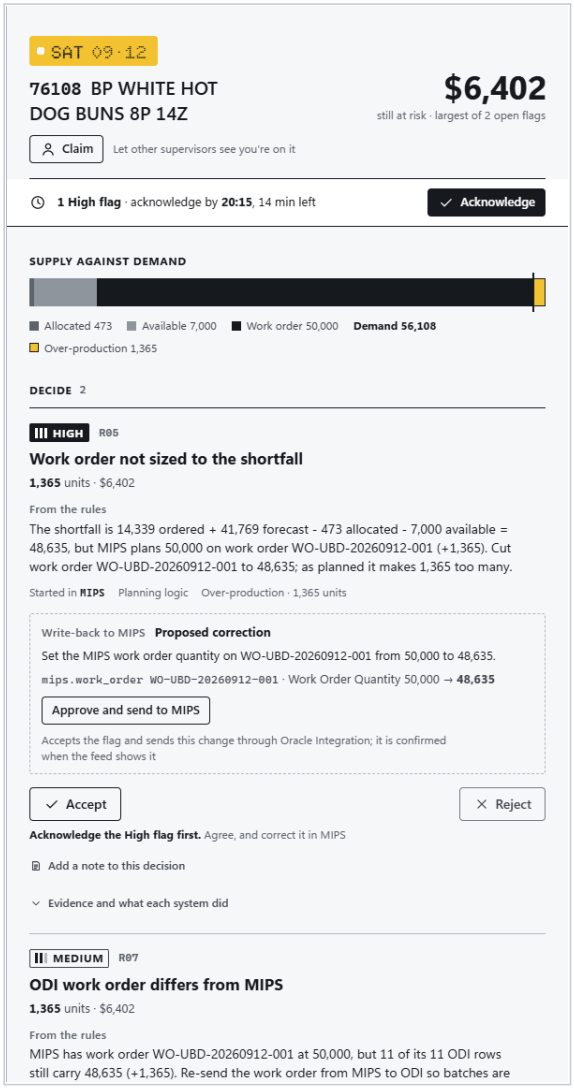
Scenario 5: MIPS re-plans the work order; ODI keeps the old quantity

Story. A planner raises 76108's work order by 1,365 units in MIPS; ODI keeps the old quantity.

Expected flags. R05 (work order not sized to the shortfall), R07 (ODI work order differs from MIPS): 1,365 units.

Result: PASS. R05-76108 (1,365 u $6,402), R07-WO-UBD-20260912-001 (1,365 u $6,402)





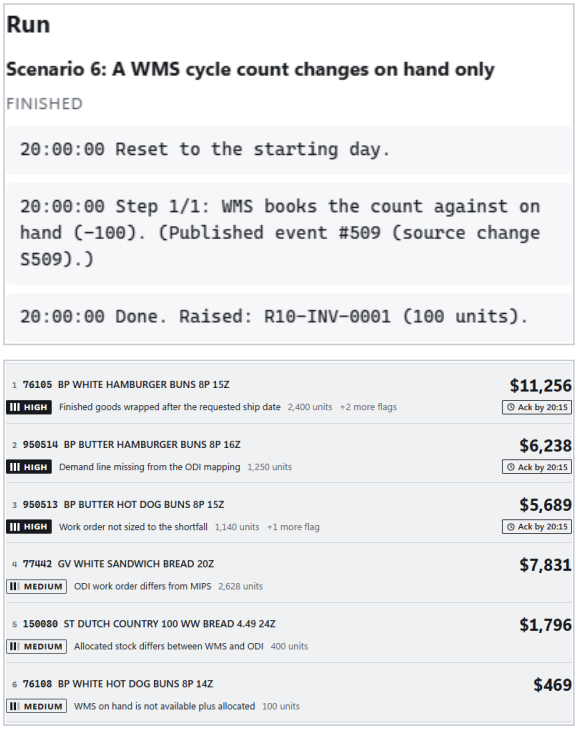
Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

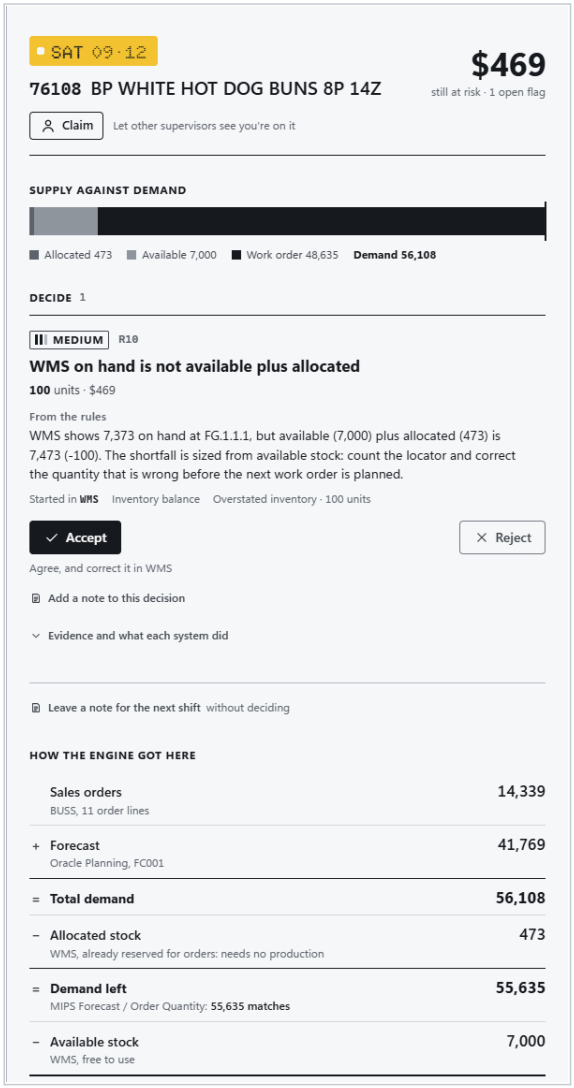
Scenario 6: A WMS cycle count changes on hand only

Story. A cycle count books 100 fewer buns against on hand only.

Expected flags. R10 (on hand is not available + allocated): 100 units.

Result: PASS. R10-INV-0001 (100 u $469)





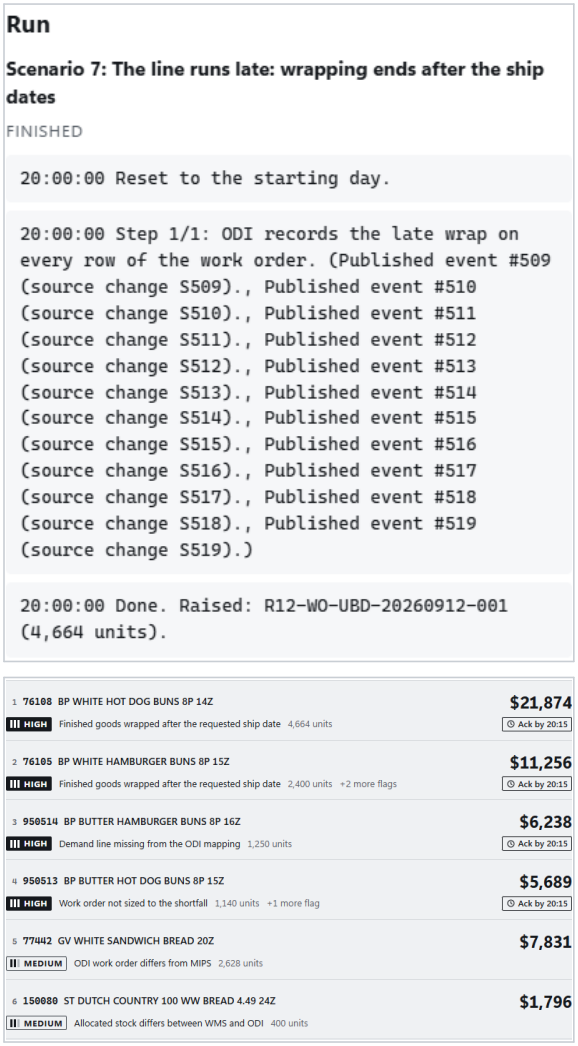
Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

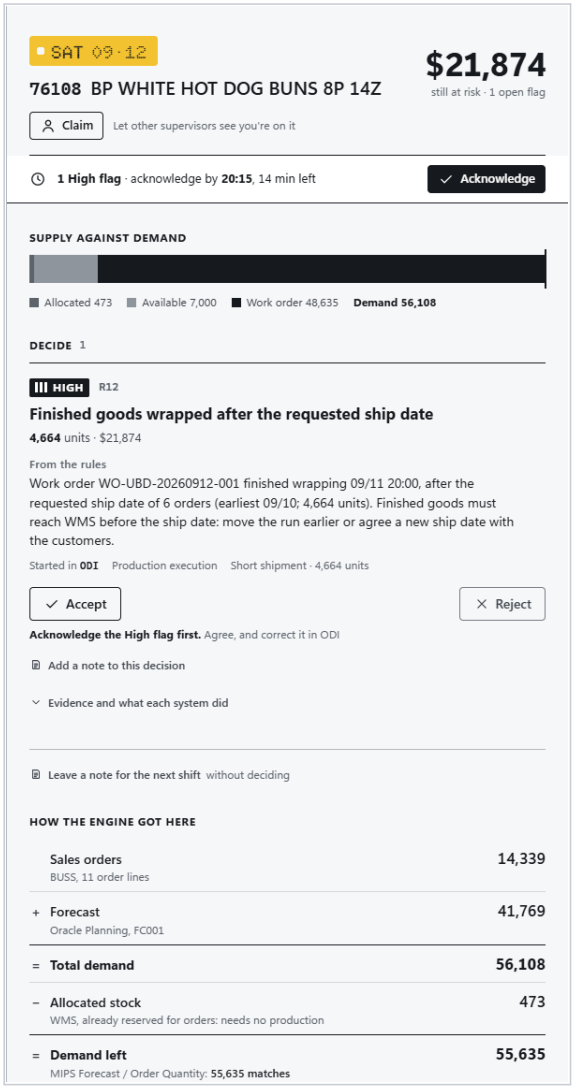
Scenario 7: The line runs late: wrapping ends after the ship dates

Story. Line 1 goes down; 76108's work order wraps after the requested ship dates.

Expected flags. R12 (finished goods wrapped after the ship date): 4,664 units.

Result: PASS. R12-WO-UBD-20260912-001 (4,664 u $21,874)





Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

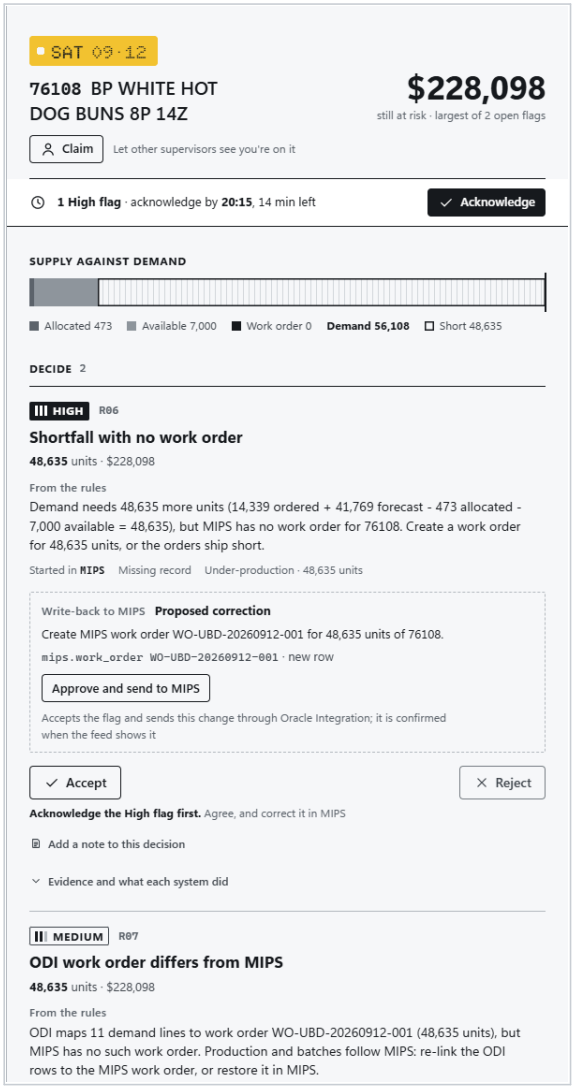
Scenario 8: The work order is deleted in MIPS

Story. 76108's work order is deleted in MIPS by mistake; ODI still maps the demand to it.

Expected flags. R06 (shortfall with no work order), R07: 48,635 units.

Result: PASS. R06-76108 (48,635 u $228,098), R07-WO-UBD-20260912-001 (48,635 u $228,098)





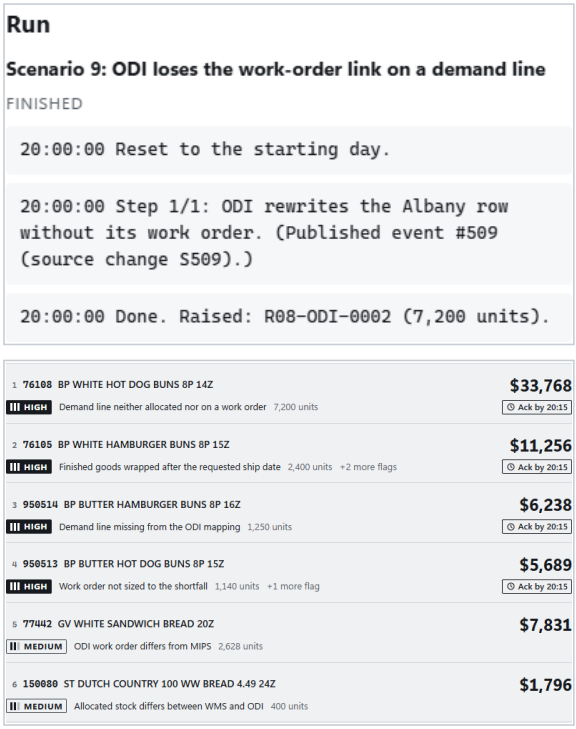
Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

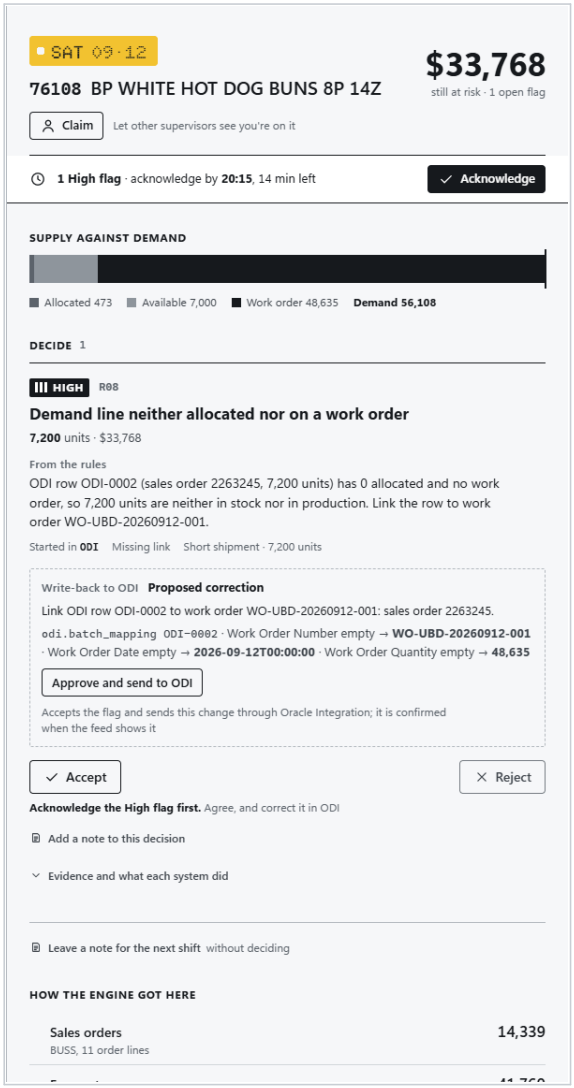
Scenario 9: ODI loses the work-order link on a demand line

Story. An ODI reload drops the work order from ALBANY NY BAKERY's line; the line has no allocated stock.

Expected flags. R08 (demand line neither allocated nor on a work order): 7,200 units.

Result: PASS. R08-ODI-0002 (7,200 u $33,768)





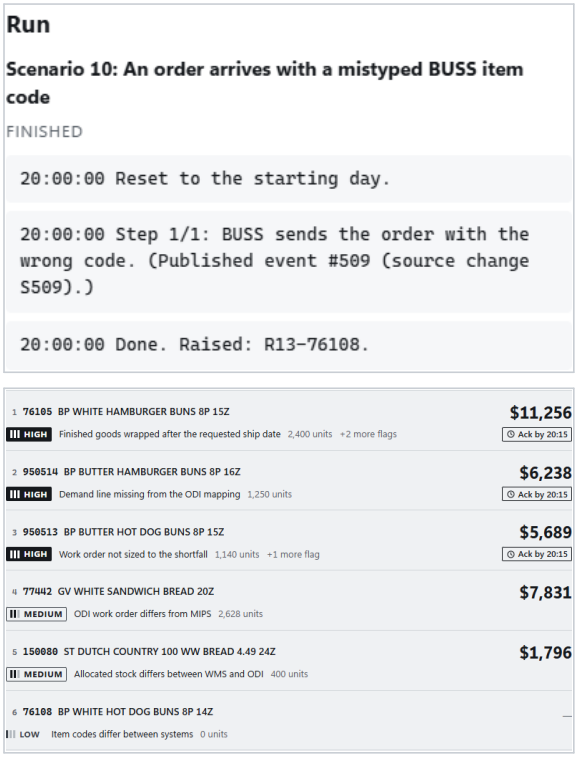
Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

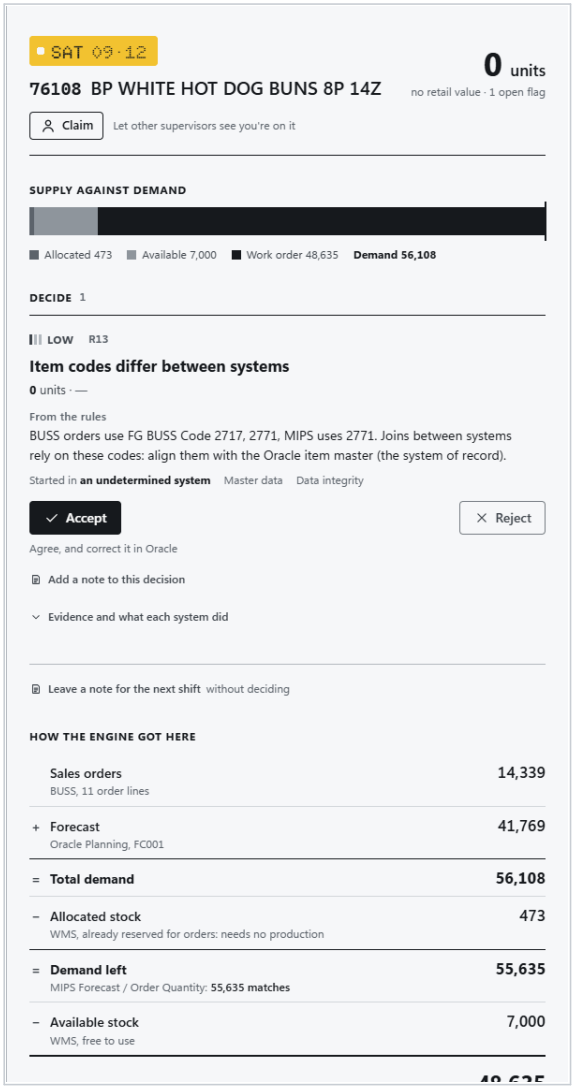
Scenario 10: An order arrives with a mistyped BUSS item code

Story. CHICOPEE MA's 76108 order arrives from BUSS with a mistyped item code (2717).

Expected flags. R13 (item codes differ between systems).

Result: PASS. R13-76108 (0 u $0)





Left: the simulator's Run panel and the worklist queue after Update. Right: the top of 76108's detail.

10. The ML layer and its scenarios

The 13 rules check that the systems agree with each other. The ML layer checks that today looks like the past: it compares each value with the same item's earlier sales days and flags what lies far outside its normal range. It catches the mistakes every system agrees on.

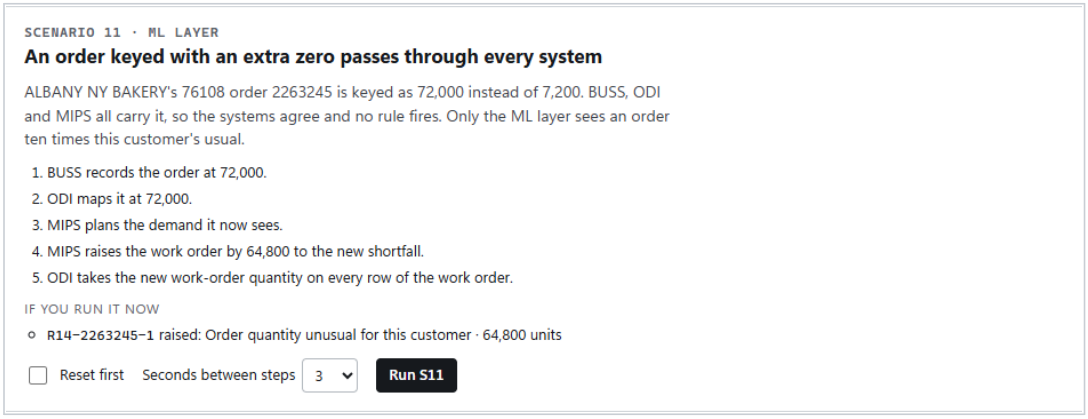
Flag	Detector	Compares today's...	...with
R14	Unusual order quantity	BUSS order line	the same customer's earlier orders of the item
R15	Forecast bias	Oracle forecast	the item's earlier forecasts
R16	Production anomaly	wrap duration of the work order (ODI)	the item's earlier work orders
R17	Inventory drift	latest change in WMS on hand	that stock row's earlier changes

How it decides

- A **robust z-score**: how far today's value is from the median of its history, measured in units of the median absolute deviation (MAD). Median and MAD ignore odd values, so one earlier outlier cannot hide the next.
- A value is flagged at a z-score of 3.5 or more either way, and is **High** at 7 or more.
- At least **6 earlier values** are needed. Without history (for example the business team's one-day workbook) the ML layer raises nothing.
- History: earlier sales days in the same tables (the demo seeds 8 earlier Saturdays). WMS rows have no date, so their history is the stock row's change log in the feed (one count a day in the demo).

What an ML flag looks like

ML flags work like every other flag: ranked, acknowledged, decided, explained by the AI when it arrives. Their explanation is labelled **From the ML layer**, the evidence shows the method, the value, the history median, the robust z and the number of history points, and the accountability names the system the value entered in as the **likely** origin (failure type: Statistical anomaly). There is no automatic correction: an unusual value is not proof of an error, so a person confirms it first.



An ML scenario card in the simulator, labelled 'ML layer'.

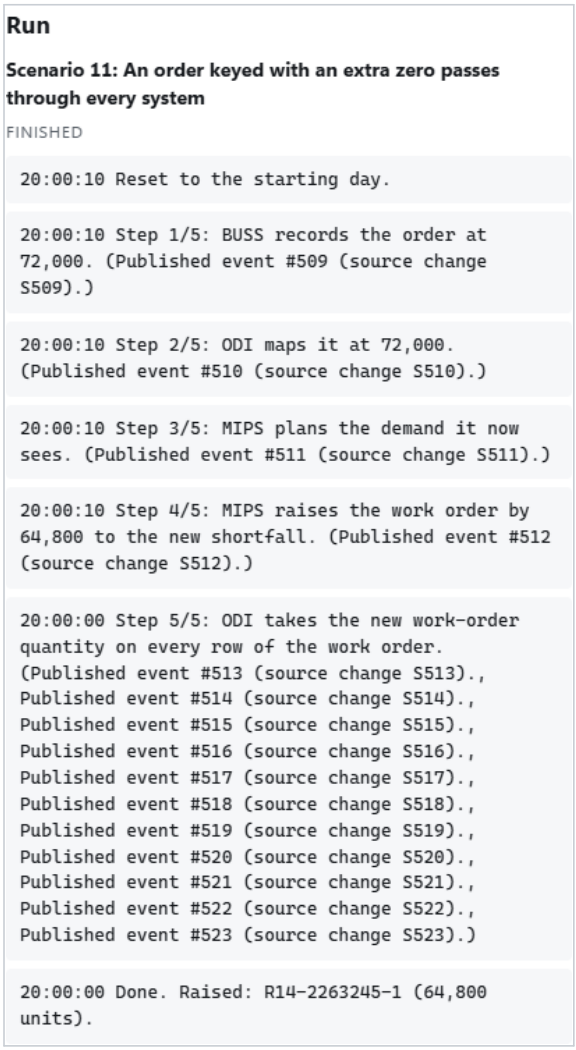
Each ML scenario keeps every system consistent, so no rule fires and only the ML flag shows the problem. Run them like the rule scenarios: tick Reset first, click Run, then Update in the worklist and open 76108.

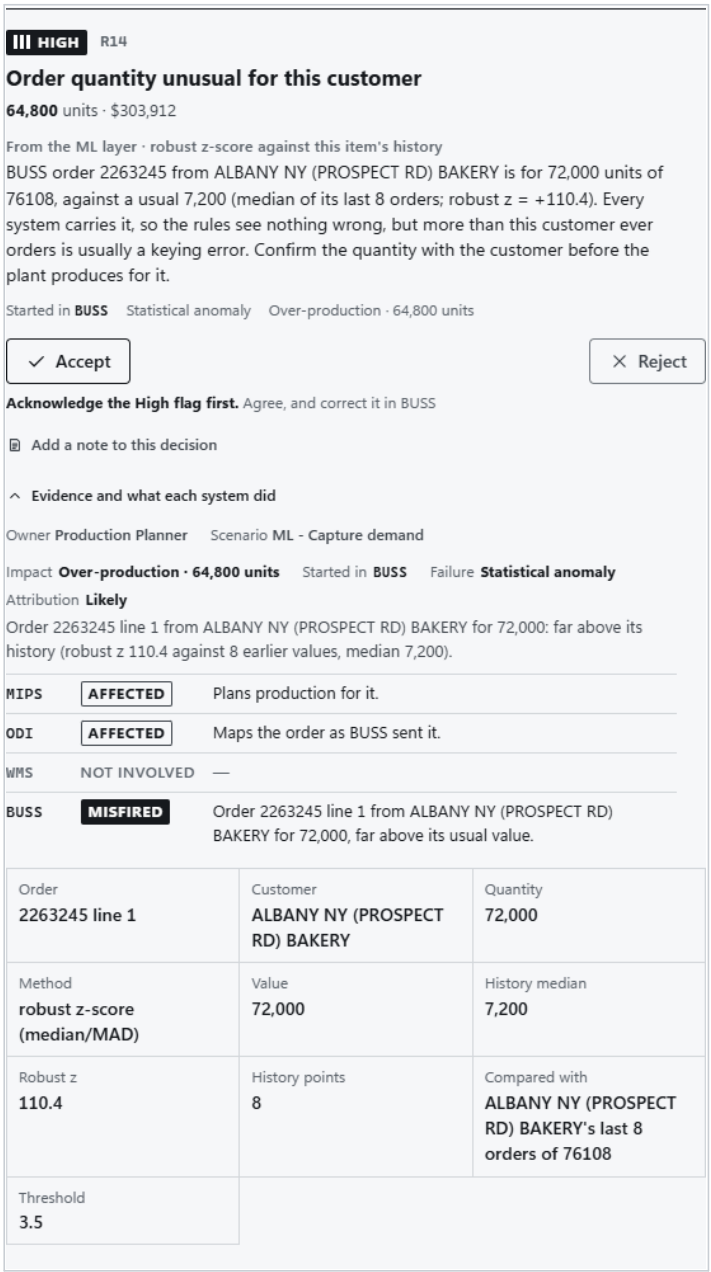
Scenario 11: Unusual order quantity (R14)

Story. ALBANY NY BAKERY's 76108 order 2263245 is keyed as 72,000 instead of 7,200. BUSS, ODI and MIPS all carry it, so the systems agree and no rule fires.

Expected. R14 only: this customer's last 8 orders of 76108 were all about 7,200.

Result: PASS. Flags on 76108 after Update: ['R14-2263245-1']





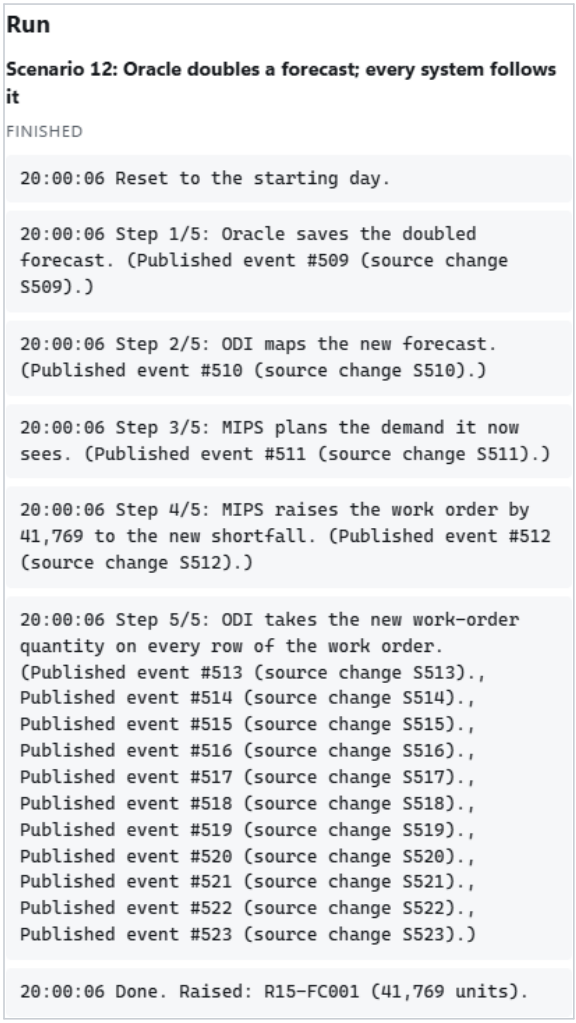
Left: the simulator's Run panel. Right: the flag in the worklist with its evidence open.

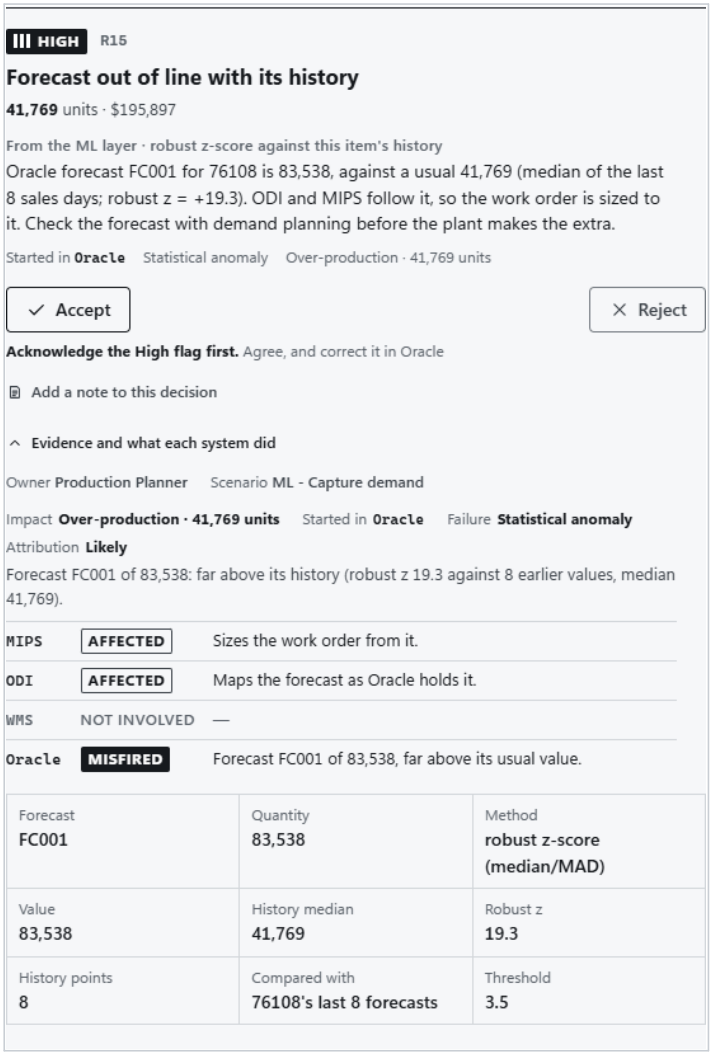
Scenario 12: Forecast bias (R15)

Story. A planning run doubles 76108's Oracle forecast FC001 from 41,769 to 83,538. ODI and MIPS follow it.

Expected. R15 only: the last 8 Saturdays' forecasts were all about 41,800.

Result: PASS. Flags on 76108 after Update: ['R15-FC001']





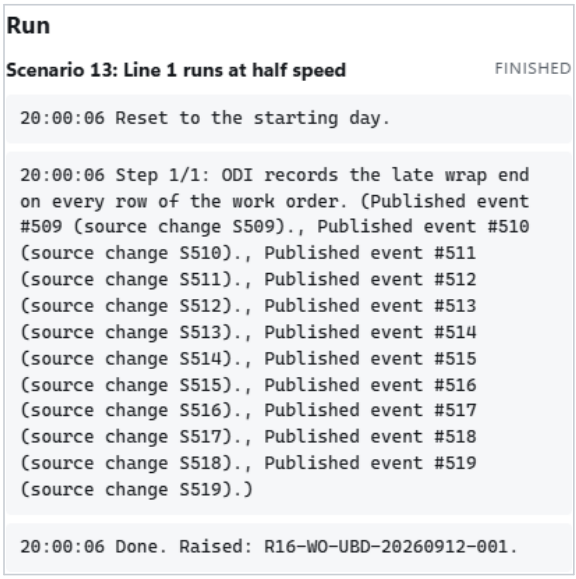
Left: the simulator's Run panel. Right: the flag in the worklist with its evidence open.

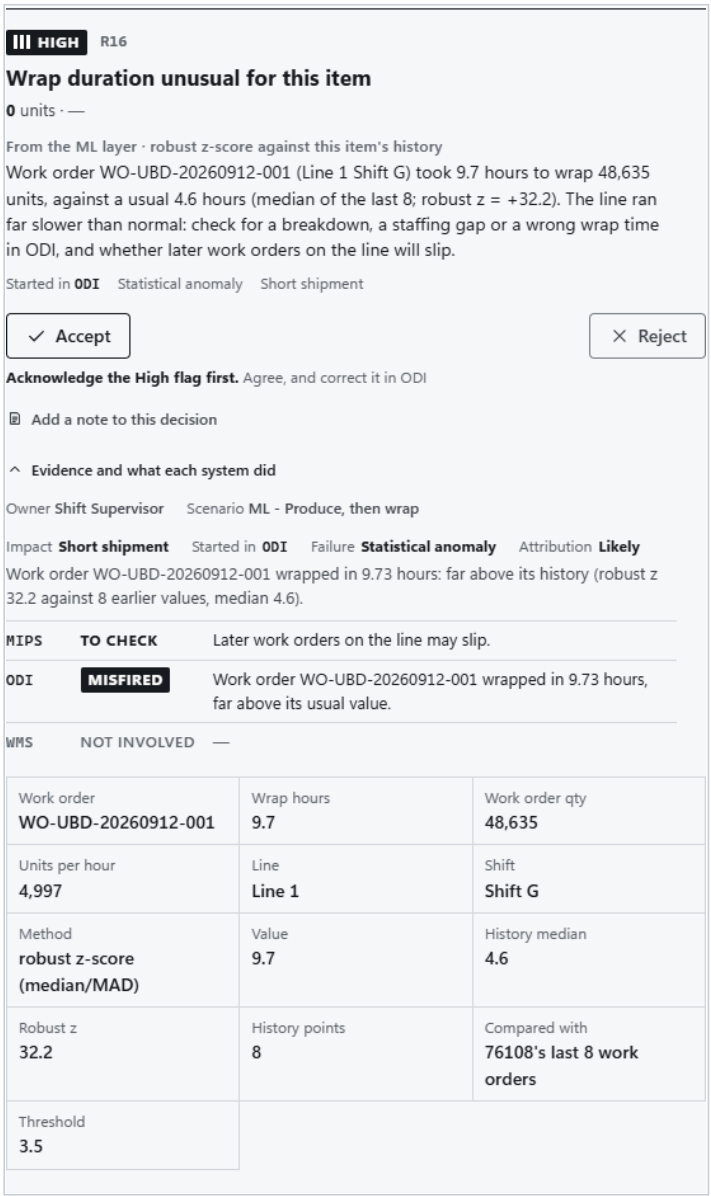
Scenario 13: Production anomaly (R16)

Story. Line 1 runs at half speed: 76108's work order takes 9 h 44 min to wrap instead of 4 h 34 min. Wrapping still ends before every requested ship date.

Expected. R16 only: the last 8 work orders wrapped in about 4.6 hours.

Result: PASS. Flags on 76108 after Update: ['R16-WO-UBD-20260912-001']





Left: the simulator's Run panel. Right: the flag in the worklist with its evidence open.

Scenario 14: Inventory drift (R17)

Story. WMS books 3,000 fewer 76108 buns on hand and available overnight, with no shipment or production, and MIPS re-plans the work order. Every system balances.

Expected. R17 only: the last 8 daily changes of this stock row moved it by about 150 units.

Result: PASS. Flags on 76108 after Update: ['R17-INV-0001']

Run

Scenario 14: Stock vanishes overnight; the books still balance

FINISHED

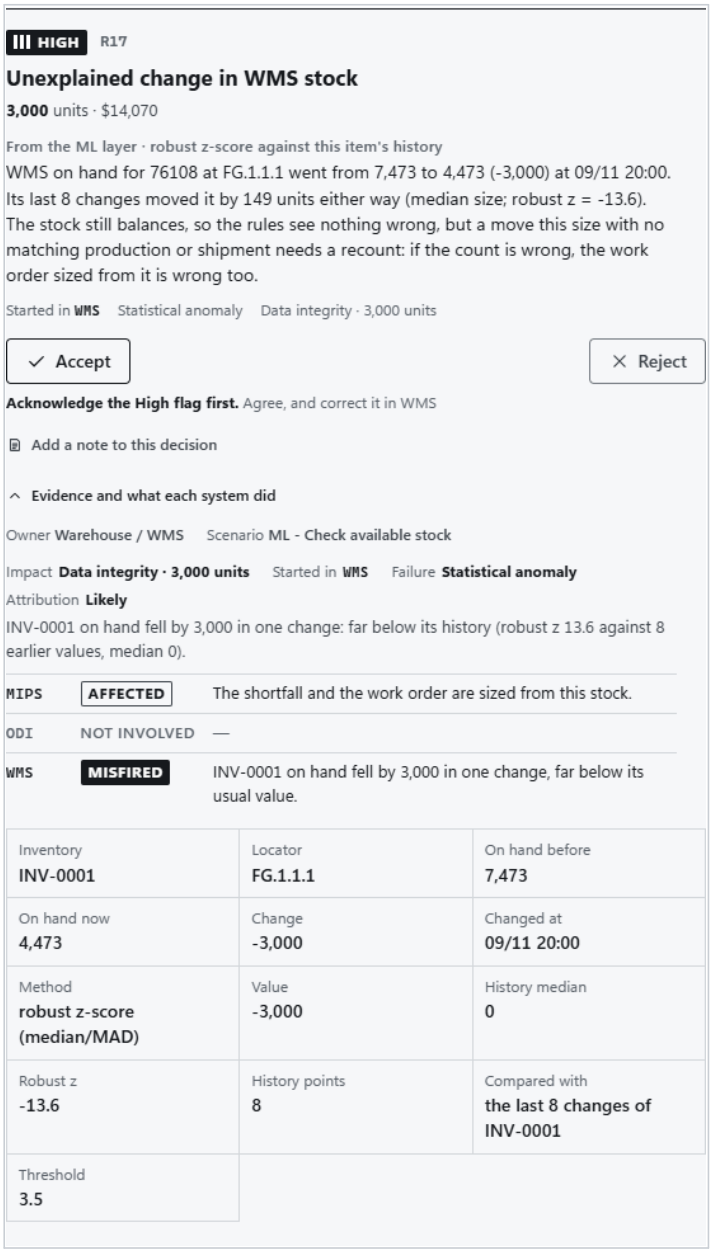
```
20:00:06 Reset to the starting day.

20:00:06 Step 1/3: WMS books 3,000 fewer on hand
and available. (Published event #509 (source
change S509).)

20:00:06 Step 2/3: MIPS raises the work order by
3,000 for the bigger shortfall. (Published event
#510 (source change S510).)

20:00:06 Step 3/3: ODI takes the new work-order
quantity on every row of the work order.
(Published event #511 (source change S511).,
Published event #512 (source change S512).,
Published event #513 (source change S513).,
Published event #514 (source change S514).,
Published event #515 (source change S515).,
Published event #516 (source change S516).,
Published event #517 (source change S517).,
Published event #518 (source change S518).,
Published event #519 (source change S519).,
Published event #520 (source change S520).,
Published event #521 (source change S521).)

20:00:06 Done. Raised: R17-INV-0001 (3,000 units).
```



Left: the simulator's Run panel. Right: the flag in the worklist with its evidence open.

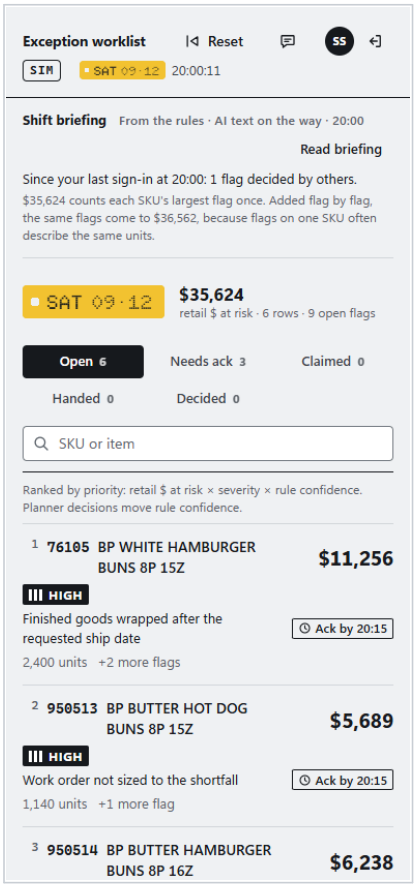
11. Rule reference

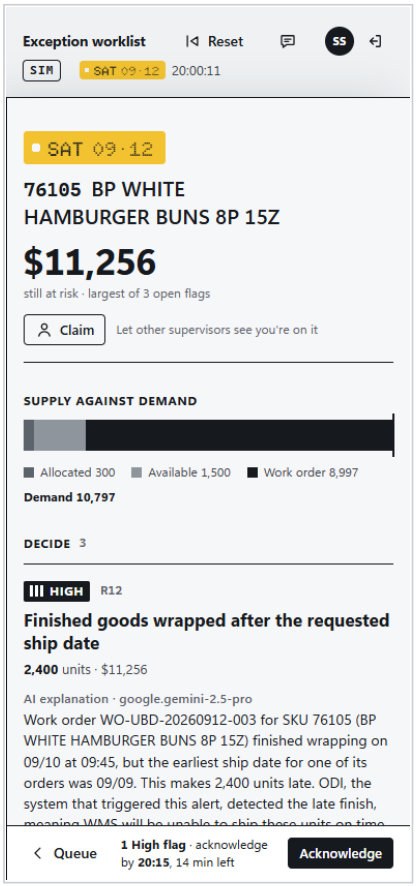
Thirteen rules, one per step of the demand-to-delivery flow, and four ML detectors (R14-R17). **Severity** is High, Medium or Low, mostly by the units involved. **Priority** = retail dollars at risk × severity × rule confidence (planner decisions move confidence). Retail dollars use the BUSS price ($4.69 for 76108).

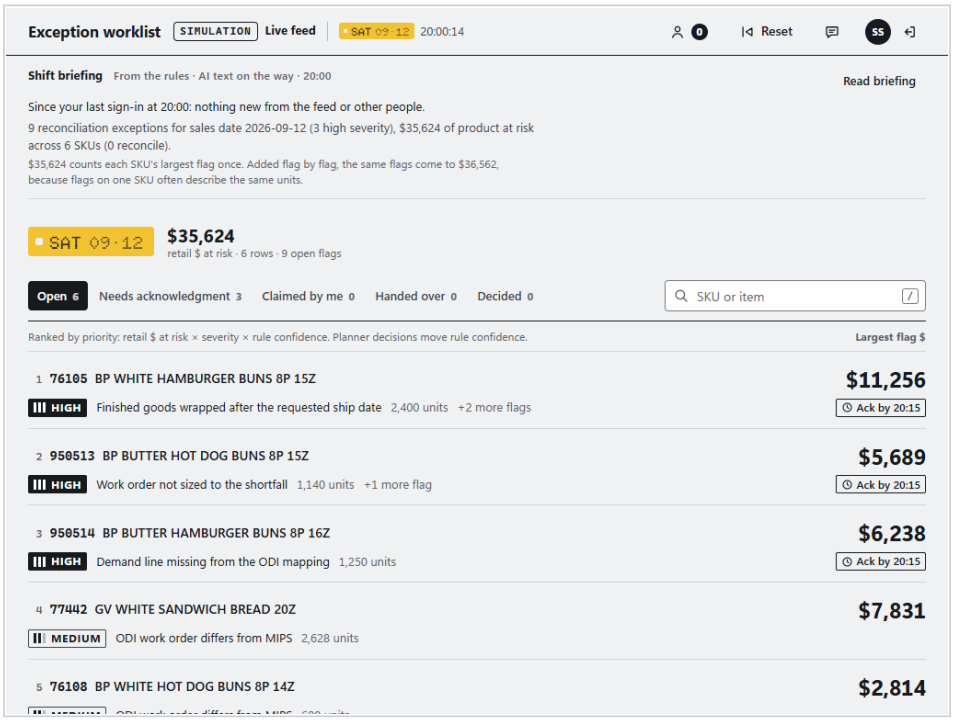
Rule	Step	Flags when	Correction proposed
R01	Capture demand	A BUSS order or Oracle forecast is missing from the ODI mapping.	Insert the ODI row
R02	Capture demand	ODI's quantity differs from the BUSS order or Oracle forecast.	Set the ODI quantity
R03	Capture demand	An ODI row maps an order or forecast that no longer exists.	Delete the ODI row
R04	Remove allocated stock	MIPS demand is not total demand less allocated stock.	Set the MIPS demand
R05	Shortfall to work order	The MIPS work order is not the shortfall (over- or under-production).	Set the work order quantity
R06	Create work order	There is a shortfall but no MIPS work order.	Create the work order
R07	Work order to ODI	ODI's work order (number or quantity) differs from MIPS.	Set every ODI row of it
R08	Allocate or produce	A demand line has neither allocated stock nor a work order.	Link it to the work order
R09	Remove allocated stock	ODI's allocated quantity differs from WMS.	No: a person decides
R10	Check stock	WMS on hand is not available + allocated.	No: a count decides
R11	Produce	Batches mixed differ from the MIPS schedule.	No
R12	Move finished goods to WMS	Finished goods were wrapped after the requested ship date.	No
R13	Master data	The item or BUSS code differs between systems.	No
R14	ML: capture demand	An order is far outside this customer's earlier orders of the item.	No: confirm it
R15	ML: capture demand	The forecast is far outside the item's earlier forecasts.	No: confirm it
R16	ML: produce, then wrap	The wrap duration is far outside the item's earlier work orders.	No: confirm it
R17	ML: check stock	A change in WMS on hand is far outside the row's earlier changes.	No: recount

12. Phones and tablets

The worklist works on any screen. On a phone the queue fills the screen; a SKU opens full screen with **Queue** to go back and the acknowledgment bar fixed at the bottom; the top bar uses two lines. On a tablet the detail slides over the queue, and the account and assistant become icons.





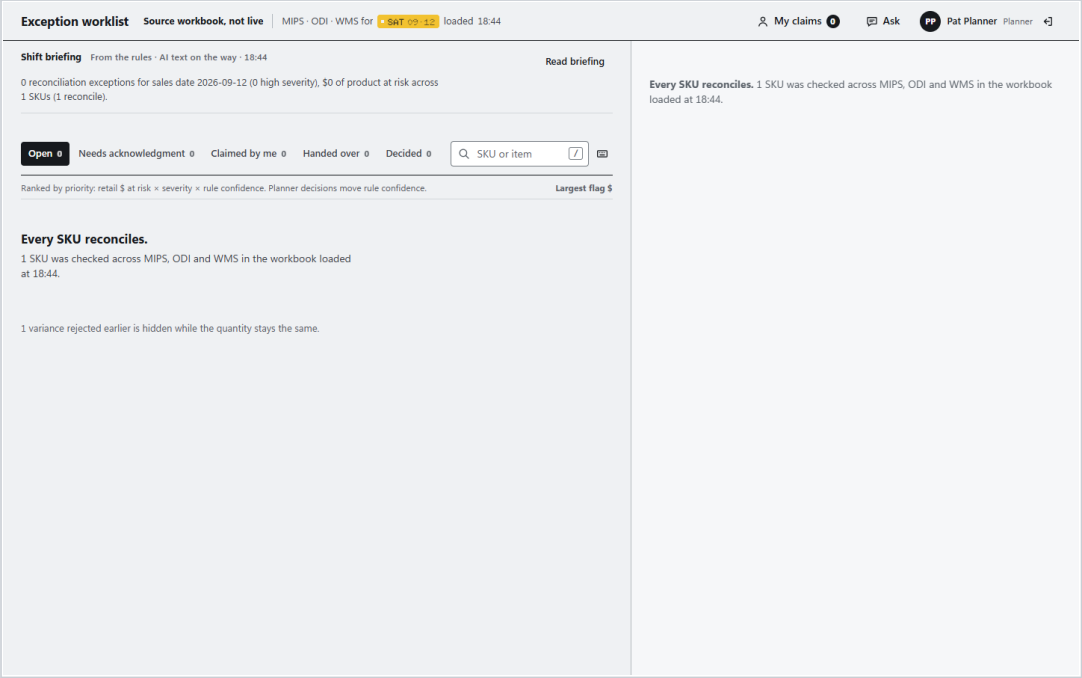


Tablet (1024 px).

13. Workbook mode, the command line and the API

13.1 Workbook mode

Without the live feed, the worklist reads the business workbook (BIMBO_MIPS_WMS_ODI_Data*.xlsx anywhere under data/). The top bar says **Source workbook, not live** with the load time, and the worklist reloads when the file changes. On the 9 Nov workbook, 76108 reconciles, so the queue says every SKU reconciles.



Workbook mode on the business workbook.

13.2 Command line

Command	What it does
python -m recon_agent --version	Shows the version (0.5.0).
python -m recon_agent serve [--port 8000]	Runs the worklist.
python -m recon_agent simulate [--port 8100]	Runs the source simulator.
python -m recon_agent run --offline [--out-dir DIR]	Reconciles the workbook and writes the planner workbook (.xlsx), the dashboard (.html) and the data (.json).
python -m recon_agent ask "question"	Asks the assistant from the command line.
python -m recon_agent users list | seed | add | passwd | role | remove	Manages accounts.
python -m recon_agent events seed --reset	Publishes the starting day into an empty event log.
python -m recon_agent events status | tail | show ENTITY [KEY]	Inspects the log and the rows.
python -m recon_agent events set ENTITY KEY "Column=value"	Changes a row (also insert, delete).
python -m recon_agent events scenario list | run S1 [--reset] [--pause N]	Lists or runs scenarios.

Command	What it does
python -m recon_agent events clock [--advance 15m | --rate N | --pause | --resume]	Controls the clock.
python -m recon_agent events fault ... / deliver	Injects integration faults; delivers delayed changes.
python -m recon_agent events export [--out DIR]	Writes the Excel, SQLite and Postgres exports.
python -m recon_agent events entities	Prints the event contract.

13.3 The Reconciliation API

All /api endpoints need a signed-in session and return JSON; /healthz is public.

Endpoint	Returns
GET /healthz	Status, version and data source.
GET /api/flags[?status=all]	Open (or all) flags with evidence, explanation and accountability.
GET /api/positions/{sku}	The reconciled position and the ledger for a SKU.
GET /api/briefing/{shift}	The briefing and top actions.
GET /api/feed	Feed mode, event number, open rows, duplicates and stale events ignored.
POST /api/flags/{id}/decision	Records accept or reject (with a note).
POST /api/corrections · GET /api/corrections[/{id}]	Approves a correction (planner or manager); lists corrections with their history.
POST /api/ask	Asks the assistant.

14. Test report

A full user test was run on 10 November 2026 against version 0.5.0: both applications on a private copy of the starting day, four accounts (planner, supervisor, manager, operator) in separate browsers, AI text from google.gemini-2.5-pro on OCI. **78 hands-on checks**: **75 passed**, **3 failed** (the known issues below). The **141 automated tests** all passed, including the ML layer's own tests.

Area	Checks	Passed
Sign-in	5	5
Worklist	13	13
AI	2	2
Working together	7	7
Decisions	5	5
Write-back	6	4
Simulator	11	10
Integration	5	5
Clock	1	1
Scenarios	11	11
ML scenarios	4	4
Layout	1	1
API	1	1
Command line	5	5
Workbook mode	1	1

Issues found

#	Issue	Effect	Workaround
1	A failed correction hides its status and reason (E4, E5).	While a correction is Failed (Oracle Integration rejected it or was unavailable), the detail shows a plain 'Proposed correction' with Retry: no Failed label and no reason.	Retry works; once retried and sent, the history lists each failure and its reason. GET /api/corrections shows them at any time.
2	The simulator's 'Reset to starting day' keeps the worklist's decisions (K1).	Flags decided before the reset come back as 'Accepted - awaiting correction' (4 of 5 in the test) and never close; open screens get no reset notice.	Reset from the worklist's 'Reset simulation' button, which clears everything.
3	The assistant can invent figures (seen in the 0.4.0 run, check O6).	Asked the same question twice from the command line, one answer quoted a work order and quantities that do not exist; the other was correct (and so was this run's). Flag explanations are figure-checked against the engine; assistant answers are not.	Check the answer against the SKU detail; ask again if a figure looks wrong.
4	Minor: /api/briefing says 'offline - set the OCI_* credentials' while the AI text is only pending.	Misleading message in the API only.	None needed.
5	Minor: workbook mode says 'MIPS · ODI · WMS' and 'checked across MIPS, ODI and WMS'.	Wording from the old data model; BUSS and Oracle are checked too.	None needed.

Every hands-on check

ID	Check	Steps	Expected	Observed	Result
A1	A signed-out visitor is sent to sign-in	Open the worklist address without signing in.	The sign-in page opens.	Redirected to /signin	**PASS**
A2	A wrong password is refused	Sign in as pat with a wrong password.	The page says the username and password do not match.	That username and password do not match.	**PASS**
A3	A planner signs in	Sign in as pat with the right password.	The worklist opens with the person's name and role in the top bar.	PP Pat Planner Planner	**PASS**
A4	The source simulator asks for the same accounts	Open the simulator without signing in.	The simulator's sign-in page opens.	/signin?next=/	**PASS**
B1	The top bar shows the feed and freshness	Look at the top bar.	Simulation badge, Live feed, the sales-day tag, the event number and the as-of time; My claims, Reset simulation, Ask, the user and Sign out.	Exception worklist SIMULATIONLive feed SAT 09·12 Sales day Saturday 12 September 2026 event #508 · as of 20:00:20 My claims 0 Reset simulation Ask PP Pat Planner Planner	**PASS**
B2	The shift briefing summarises the day	Read the briefing at the top of the queue.	The day's exceptions and the $32,810 at risk, from the rules or (when it arrives) the AI briefing.	Shift briefing From the rules · AI text on the way · 20:00 Read briefing Since your last sign-in at 20:00: nothing new from the feed or other people. 8 reconciliation exceptions for sales date 2026-09-12 (3 high severity), $32,810 of product at risk across 6 SKUs (1 reconcile). $32,810 counts each S	**PASS**
B3	Open the full briefing and mark it read	Click Read briefing, then Mark read.	The full briefing opens (top actions per SKU); Mark read records it.	Shift briefing From the rules · AI text on the way · 20:00 Mark read Since your last sign-in at 20:00: nothing new from the feed or other people. 8 reconciliation exceptions for sales date 2026-09-12 (3 high severity), $32,810 of product at risk across 6 SKUs (1 reconcile). Work orders over-produce	**PASS**
B4	The sales-day shelf shows the money at risk	Look at the day tile above the queue.	SAT 09-12: $32,810 retail at risk, 5 rows, 8 open flags.	SAT 09·12 Sales day Saturday 12 September 2026 $32,810 retail $ at risk · 5 rows · 8 open flags	**PASS**
B5	The queue is ranked by priority	Read the queue.	Five SKUs: 76105 ($11,256, High), 950514 ($6,238, High), 950513 ($5,689, High), 77442 ($7,831, Medium), 150080 ($1,796, Medium); 76108 reconciles and is not listed.	Rank 1 76105 BP WHITE HAMBURGER BUNS 8P 15Z $11,256 HIGH Finished goods wrapped after the requested ship date 2,400 units +2 more flags : R02 ODI quantity differs from the order or forecast; R10 WMS on hand is not available plus allocated Ack by 20:15 | Rank 2 950514 BP BUTTER HAMBURGER BUNS 8P 16Z …	**PASS**
B6	The view tabs filter the queue	Click Needs acknowledgment, Claimed by me, Handed over, Decided, then Open.	Needs acknowledgment lists both High SKUs; the empty views explain what would appear there.	Needs acknowledgment: Rank 1 76105 BP WHITE HAMBURGER BUNS 8P 15Z $11,256 HIGH Finished goods wrapped after the requested ship date 2,400 units +2 more flags : R0; Claimed by me: You haven't claimed anything. Claim a SKU from its detail so other supervisors can see you're working it.; Handed over: N…	**PASS**
B7	Search by SKU or item name	Type 950513, then hamburger, then zzz in the search box.	950513 finds that SKU; hamburger finds 76105; zzz explains there is no match.	950513: Rank 3 950513 BP BUTTER HOT DOG BUNS 8P 15Z $5,689 HIGH Work order not sized to the shortfall 1,140 units +1 m; hamburger: Rank 1 76105 BP WHITE HAMBURGER BUNS 8P 15Z $11,256 HIGH Finished goods wrapped after the requested ship date ; zzz: No SKU or item matches “zzz”. Search by SKU number o…	**PASS**
B8	Keyboard shortcuts	Press ?, then Esc; press j twice and Enter.	? lists the shortcuts; j moves down the queue and Enter opens the SKU.	shortcut list shown: True; opened by keyboard: 950514	**PASS**
B9	The SKU detail explains the flag end to end	Open 950513.	Header with $ at risk, Claim, the High-flag acknowledgment bar, the supply-against-demand bar, each flag with its rule text and a proposed write-back, the engine's working (ledger), each system's quantity and the source records.	header ok, claim ok, ack bar ok, track ok, R05 explanation ok, write-back ok, R11 ok, ledger ok, systems ok, sources ok	**PASS**

ID	Check	Steps	Expected	Observed	Result
B10	Evidence and system accountability for a flag	On 950513's R05, open 'Evidence and what each system did'.	The evidence values and each system's role (Misfired, Reference, Affected, To check), with where it started.	HIGH R05 Work order not sized to the shortfall 1,140 units · $5,689 From the rules The shortfall is 2,200 ordered + 3,000 forecast - 360 allocated - 1,140 available = 3,700, but MIPS plans 4,840 on work order WO-UBD-20260912-002 (+1,140). MIPS sized the work order to the 4,840 demand left and did no…	**PASS**
B11	Source records behind the position	Open 'Source records' at the bottom of the detail.	The BUSS orders, the forecast, the ODI rows, the WMS stock and the MIPS work order, as each system holds them.	SOURCE RECORDS 3 BUSS orders · 1 forecast · 4 ODI rows · 1 WMS · 1 MIPS BUSS sales orders Order Customer Ship date Qty 2263246 ALBANY NY (PROSPECT RD) BAKERY 09/12 1,200 2263258 CHICOPEE MA DISTRIBUTION CENTER 09/12 360 2266621 CANASTOTA NY DISTRIBUTION CENTER 09/12 640 Oracle forecast Forecast Date	**PASS**
B12	The starting day raises exactly the designed flags	Read the open flags.	R12-WO-UBD-20260912-003 2,400 u $11,256; R01-2263271-1 1,250 u $6,238; R05-950513 1,140 u $5,689; R07-WO-UBD-20260912-006 2,628 u $7,831; R09-150080 400 u $1,796; R02-2263255-1 100 u $469; R10-INV-0003 100 u $469; R11-WO-UBD-20260912-002 0 u.	R12-WO-UBD-20260912-003 2,400 u $11,256; R01-2263271-1 1,250 u $6,238; R05-950513 -1,140 u $5,689; R07-WO-UBD-20260912-006 2,628 u $7,831; R09-150080 400 u $1,796; R02-2263255-1 100 u $469; R10-INV-0003 100 u $469; R11-WO-UBD-20260912-002 0 u $0	**PASS**
B13	AI explanations replace the rule text when they arrive	Wait for the AI text (about 20 s after the data settles); open 950513's R05 evidence.	The briefing and each flag switch to the AI explanation, labelled with the model; the evidence shows the AI's cause chain, a check that every figure matches the engine, and an audit id.	briefing source: AI briefing · google.gemini-2.5-pro · 20:00; flag: HIGH R05 Work order not sized to the shortfall 1,140 units · $5,689 AI explanation · google.gemini-2.5-pro MIPS planned to overproduce SKU 950513 by 1,140 units. The correct production shortfall is 3,700 units, but MIPS created work…	**PASS**
B14	Ask the assistant a question	Click Ask, type a question about 950513 and send it.	An answer grounded in the current data (figures from the engine), shown in the assistant panel.	Why does 950513 have a work order problem, and what should I change? AI answer · google.gemini-2.5-pro on OCI · numbers from the Reconciliation API For SKU 950513, Ball Park Butter Hot Dog Buns, the MIPS work order is not sized correctly to the shortfall. MIPS plans a work order for 4,840 units, but	**PASS**
C1	Claim a SKU; others see it	Pat opens 950513 and clicks Claim. Sam (supervisor, another browser) clicks Update.	Pat's 'My claims' becomes 1. Sam's top bar shows the change; after Update, Sam sees Pat's claim and a warning that deciding records Sam's name, not Pat's.	Pat my claims: My claims 1; Sam's pill: 1 row changed by others · Update; Sam sees: claimed by Pat	**PASS**
C2	Claimed by me	Pat clicks the Claimed by me tab.	Only 950513 is listed.	Rank 3 950513 BP BUTTER HOT DOG BUNS 8P 15Z $5,689 HIGH Work order not sized to the shortfall 1,140 units +1 more flag :	**PASS**
C3	Acknowledge a High flag	Pat opens 76105 and clicks Acknowledge.	The High flag is acknowledged with Pat's name and time; Needs acknowledgment drops by one.	needs ack 3 -> 2; detail says: acknowledged	**PASS**
C4	Leave a shift note; Handed over collects touched work	Sam writes a note on 76105 ('Leave a note for the next shift'). Pat clicks Update, then Handed over.	The note shows on 76105 with Sam's name and time. Handed over lists 76105 (note, acknowledged High flag) and 950513 is not there for Pat (Pat's own claim).	Pat's pill: 1 row changed by others · Update; Handed over: Rank 1 76105 BP WHITE HAMBURGER BUNS 8P 15Z $11,256 HIGH Finished goods wrapped after the requested ship date 2,400 units +2 more flags : R02 ODI quantity diffe	**PASS**
C5	Release a claim	Pat clicks Release on 950513.	The claim is gone and My claims returns to 0.	my claims: My claims 0	**PASS**
C6	What changed since your last sign-in	Pat signs in again (a new browser).	The briefing says what others did since Pat's previous sign-in (Sam's shift note) and links Handed over.	Since your last sign-in at 20:00: 1 shift note. Handed over 1	**PASS**
C7	An unacknowledged High flag goes overdue	In the simulator, click +15 min on the plant clock; reload the worklist.	950513's High flag is still unacknowledged, so its 'Ack by 20:15' turns overdue.	Rank 1 76105 BP WHITE HAMBURGER BUNS 8P 15Z $11,256 HIGH PP Acknowledged by Pat Planner, not decided yet 1 shift note Rank 2 950514 BP BUTTER HAMBURGER BUNS 8P 16Z $6,238 HIGH Ack overdue 3 min Rank 3 950513 BP BUTTER HO	**PASS**

ID	Check	Steps	Expected	Observed	Result
D1	Accept a flag with a note	Pat acknowledges 950513's High flag, adds a note to R11 and clicks Accept.	R11 shows Accepted by Pat with the time and the note; the day shelf counts the decision; the detail says how the row's rank moved.	flag: LOW R11 Batches mixed differ from the MIPS schedule 0 units · — No retail value AI explanation · google.gemini-2.5-pro The production floor mixed more batches than scheduled for work order WO-UBD-20260912-002. MIPS sched | shelf: SAT 09·12 Sales day Saturday 12 September 2026 $32,810 retail $ ...	PASS
D2	Reject a flag	Pat clicks Reject on 76105's R10.	R10 shows Rejected; the rule's confidence drops and the flag stays hidden on the next run while its quantity is unchanged.	MEDIUM R10 WMS on hand is not available plus allocated 100 units · $469 AI explanation · google.gemini-2.5-pro WMS shows an inventory imbalance for SKU 76105 at locator FG.2.1.1. The on-hand quantity is 1,900 units, but	PASS
D3	Change a decision	Pat opens 'Change' on the rejected R10 and clicks Accept.	R10 is now Accepted.	MEDIUM R10 WMS on hand is not available plus allocated 100 units · $469 AI explanation · google.gemini-2.5-pro WMS shows an inventory imbalance for SKU 76105 at	PASS
D4	The Decided tab counts flags	Pat clicks Decided.	The tab counts 2 decided flags and lists their SKUs.	count 2; Rank 1 76105 BP WHITE HAMBURGER BUNS 8P 15Z $11,256 HIGH Finished goods wrapped after the requested ship date 2,400 units +1 more flag : R02 ODI quantity differ	PASS
D5	Decisions move the money on the shelf	Look at the day tile after the decisions.	It shows what is still at risk and what was accepted and decided.	SAT 09·12 Sales day Saturday 12 September 2026 $32,810 retail $ at risk · 5 rows · 6 open flags $469 accepted (each accepted flag's own retail $, added up; flags on one SKU can overlap) · 2 flags decided	PASS
E1	Only planners and managers approve write-back	Sam (supervisor) opens 950513's R05.	The proposed correction is shown, without the Approve button: 'A planner or manager approves write-back.'	Write-back to MIPS Proposed correction Set the MIPS work order quantity on WO-UBD-20260912-002 from 4,840 to 3,700. mips.work_order WO-UBD-20260912-002 · Work Order Quantity 4,840 → 3,700 A planner or manager approves write-back.	PASS
E2	Approve and send a correction to MIPS	Pat clicks 'Approve and send to MIPS' on 950513's R05 (work order 4,840 -> 3,700).	The flag is accepted and the correction goes to Oracle Integration (the simulator); MIPS applies it, the change comes back through the feed, the correction is Confirmed and R05 closes.	COR-0001: Approved (Pat Planner) -> Sent (Corrections API) -> Confirmed (feed); R05 closed: True; MIPS work order now 3,700: True	PASS
E3	The SKU after its correction	Look at 950513 after the correction is confirmed.	R05 is gone and the supply bar shows no over-production; R11 (batches) remains.	SUPPLY AGAINST DEMAND Supply against demand on one scale: allocated 360, available 1,140, work order 3,700, demand 5,200. Allocated 360 Available 1,140 Work order 3,700 Demand 5,200	PASS
E4	Oracle Integration rejects a correction	Operator sets the corrections endpoint to 'Reject every correction'. Mona (manager) approves 76105's R02 (ODI 800 -> 900).	The correction shows Failed with the reason, and a Retry button.	DEFECT. The correction is recorded as Failed with the reason 'Oracle Integration rejected it' (GET /api/corrections), but while it is Failed the SKU detail shows only a 'Proposed correction' with Retry: no Failed status and no reason. The reasons appear in the history only after a retry is sent (E6)...	FAIL
E5	Oracle Integration is unavailable	Operator sets 'Unavailable (503)'. Mona clicks Retry.	The correction fails again, saying the endpoint is unavailable, and keeps its history.	DEFECT (same cause as E4). Recorded as Failed: 'Oracle Integration is unavailable'; Retry works, but the screen shows only 'Proposed correction' and Retry, without the failure or its reason.	FAIL
E6	Retry after recovery; a slow integration keeps it Sent	Operator sets 'Accept and apply' and adds a 120 s delay on ODI. Mona clicks Retry. Operator then clicks 'Deliver waiting now'.	The correction is accepted and shows 'Sent, waiting for the change in the feed' while the change is held; once delivered it is Confirmed and R02 closes.	while held: Write-back to ODI COR-0002 · Sent, waiting for the change in the feed Set ODI row ODI-0019 from 800 to 900, as BUSS has | integration: Active faults # Fault Applies to Left Note Remove 1 delay 120 s ODI until removed OIC agent restart Remove 1 delayed · 0 | history: Approved -> Failed -...	PASS
F1	Change a BUSS order quantity	Simulator > BUSS Sales orders > 2263245-1: Order Quantity 7,200 -> 7,800, note, Publish change.	The worklist shows '1 row changed by others · Update'. After Update, 76108 (which reconciled) appears with R02 (ODI still has 7,200), R04 (MIPS demand) and R05 (work order), each 600 units.	pill: 1 row changed by others · Update; new flags: ['R02-2263245-1', 'R04-76108', 'R05-76108']	PASS

ID	Check	Steps	Expected	Observed	Result
F2	Change WMS on hand	Simulator > WMS Inventory > INV-0001: On hand 7,473 -> 7,373.	The worklist raises R10-INV-0001 (on hand is not available + allocated), 100 units.	pill: 1 row changed by others · Update; R10-INV-0001: (-100.0, 469, 'Open')	**PASS**
F3	Change an Oracle forecast	Simulator > Oracle Forecast > FC001: 41,769 -> 42,769.	The worklist raises R02 on the forecast's ODI row (ODI still maps 41,769) and the MIPS demand and work order flags grow by 1,000.	pill: 1 row changed by others · Update; flags: ['R02-FC001', 'R04-76108', 'R05-76108']	**PASS**
F4	Insert a new order line (Copy as new row)	Open 2263256-1, click 'Copy as new row', set SO Line ID 2263299-1, Order Number 2263299, quantity 500, Insert row.	The worklist raises R01-2263299-1: the new order is missing from the ODI mapping.	pill: 1 row changed by others · Update; R01-2263299-1: (500.0, 2345, 'Open')	**PASS**
F5	Delete a row	Open 2263299-1, click Delete and confirm.	A delete event is published and R01-2263299-1 clears in the worklist.	pill: 1 row changed by others · Update; R01-2263299-1 still open: False	**PASS**
F6	A change that moves no flag still reaches the screen	Simulator > ODI Batch mapping > ODI-0013: Act Batches Mixed 5 -> 6.	The worklist still shows the Update pill; after Update 950513's R11 says the ODI rows disagree (ODI-0013: 6 of 4).	pill: 1 row changed by others · Update; R11 text mentions ODI-0013 6 of 4: True	**PASS**
F7	Two operators edit the same row: the older form is refused	Operator A opens INV-0002. Operator B changes and publishes it. Operator A publishes the old form.	Operator A's change is refused because the row changed since the form was opened.	This row changed (event #515) after you opened it. Its current values are shown; make your change again. INV-0002 WMS · wms.inventory · last change S515 Inventory ID* Item Number* Item Description On	**PASS**
F8	The Source changes log	Look at 'Source changes' under any table.	Newest first: each change with its operation, row, plant time, who made it and the note, and whether the worklist has reconciled it.	Source changes Newest first, all systems, with what the integration did with each and whether the worklist has reconciled it. S515 UPDATE wms.inventory INV-0002 event #515 · on worklist On hand 1,500 → 1,490 plant time 09/11 20:00:16 · simulator:Olly Operator · Second operator S514 UPDATE odi.batch_	**PASS**
F9	The status bar shows the integration and the worklist's progress	Look at the simulator's top bar after some changes.	Plant time and clock controls; the event log position, changes in flight, whether the worklist is up to date, the event-to-worklist latency, and Open worklist.	SIMULATION Source simulator Dummy BUSS · Oracle · MIPS · ODI · WMS data. Each change goes through the integration to the worklist. PLANT TIME Fri 09/11 20:00:10 ×1 Pause ×1 ×10 ×60 ×300 +15 min +1 h EVENT LOG #512 IN FLIGHT none WORKLIST up to date EVENT → WORKLIST p95 0.14 s · p50 0.08 s Open workl	**PASS**
G1	Stale WMS feed (delay)	Integration > 'Stale WMS feed'. Change WMS INV-0002 On hand 1,500 -> 1,400. Watch the worklist for 5 s, then click 'Deliver waiting now'.	The change waits in the integration (1 delayed) and the worklist shows nothing new; once delivered, the worklist raises R10-INV-0002.	worklist quiet while delayed: True; integration: Active faults # Fault Applies to Left Note Remove 1 delay 60 s WMS until removed added by Olly Operator Remove 1 delayed · 0 held · 0 droppe; after delivery: pill '1 row changed by others · Update', R10-INV-0002 (-100.0, 499, 'Open')	**PASS**
G2	Lose the next BUSS order (drop), then replay	Integration > 'Lose the next BUSS order'. Change BUSS 2263246-1 to 1,300. Then 'Replay dropped'.	The change never reaches the worklist (1 dropped) until it is replayed from the source; then 950513 shows R02-2263246-1 and its demand flags.	quiet while dropped: True; integration: Active faults No faults: every change reaches the worklist at once, exactly once, in order. 0 delayed · 0 held · 1 dropped · 0 duplicate cop; after replay: pill '1 row changed by others · Update', R02-2263246-1 (100.0, 499, 'Open')	**PASS**
G3	At-least-once delivery (duplicate)	Integration > 'At-least-once delivery'. Change INV-0002 On hand to 1,450.	The change arrives twice (the copy 3 s later); the worklist applies it once: R10-INV-0002 shows 50 units.	integration: emove 0 delayed · 0 held · 0 dropped · 1 duplicate copies delivered Deliver waiting now Replay dropped Remove all faults; R10-INV-0002 (-50.0, 250, 'Open')	**PASS**
G4	Out-of-order delivery	Integration > 'Out-of-order delivery'. Change 2263246-1 three times: 1,310, 1,320, 1,330.	The three changes arrive in one batch, newest first; the worklist still ends on the last value (BUSS orders for 950513 = 1,330 + 640 + 360 = 2,330).	BUSS orders for 950513 on the worklist: 2330.0	**PASS**
G5	Add a custom fault and remove it	Add a fault: drop, WMS, 1 change, note. Then Remove it.	The fault is listed with its note and how many changes it still applies to; Remove deletes it.	active: Active faults # Fault Applies to Left Note Remove 5 drop WMS 1 WMS adapter restart Remove 0 delayed · 0 held · 0 dropped · 1 duplicate copies delivere | after remove: Active faults No faults: every change reaches the worklist at once, exactly once	**PASS**

ID	Check	Steps	Expected	Observed	Result
H1	Pause, speed up and advance the plant clock	Click Pause; Resume and x60; +1 h; then x1. Reload the worklist.	Paused: plant time stands still. x60: a minute of plant time per second. +1 h jumps ahead. The worklist's as-of time and the acknowledgment deadlines follow plant time.	paused still: True; at x60: PLANT TIME Fri 09/11 20:03:34 ×60 Pause ×1 ×10 ×60 ×300 +15 min +1 h; after +1 h: PLANT TIME Fri 09/11 21:04:44 ×60 Pause ×1 ×10 ×60 ×300 +15 min +1 h; worklist: SAT 09·12 Sales day Saturday 12 September 2026 event #515 · as of 21:04:59	**PASS**
I1	S1: A customer raises a BUSS order; ODI is not refreshed	Simulator > Scenarios > S1: tick 'Reset first', 0 s between steps, Run S1. In the worklist click the top-bar pill.	New flags on 76108: R02-2263245-1, R04-76108, R05-76108.	run: Run Scenario 1: A customer raises a BUSS order; ODI is not refreshed FINISHED 21:05:09 Reset to the starting day. 21:05:09 Step 1/1: BUSS takes the bigger order (+600). (Published event #509 (source c | worklist pill: 2 rows changed by others · Update | new flags: R02-2263245-1 (600 u $2,814), ...	**PASS**
I2	S2: A new BUSS order line never reaches ODI	Simulator > Scenarios > S2: tick 'Reset first', 0 s between steps, Run S2. In the worklist click the top-bar pill.	New flags on 76108: R01-2263260-1, R04-76108, R05-76108.	run: Run Scenario 2: A new BUSS order line never reaches ODI FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: BUSS records the new order; nothing arrives in ODI. (Published event #509 (sourc | worklist pill: 6 rows changed by others · Update | new flags: R01-2263260-1 (900 u $4,221), ...	**PASS**
I3	S3: A customer cancels in BUSS; ODI keeps the line	Simulator > Scenarios > S3: tick 'Reset first', 0 s between steps, Run S3. In the worklist click the top-bar pill.	New flags on 76108: R03-ODI-0012, R04-76108, R05-76108.	run: Run Scenario 3: A customer cancels in BUSS; ODI keeps the line FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: BUSS deletes the order line. (Published event #509 (source change S509).) | worklist pill: 1 row changed by others · Update | new flags: R03-ODI-0012 (352 u $1,651), R0...	**PASS**
I4	S4: WMS allocates more stock; ODI and MIPS keep the old allocation	Simulator > Scenarios > S4: tick 'Reset first', 0 s between steps, Run S4. In the worklist click the top-bar pill.	New flags on 76108: R09-76108, R04-76108.	run: Run Scenario 4: WMS allocates more stock; ODI and MIPS keep the old allocation FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: WMS moves 660 units from available to allocated. (Publish | worklist pill: 1 row changed by others · Update | new flags: R04-76108 (660 u $3,095), R09-7...	**PASS**
I5	S5: MIPS re-plans the work order; ODI keeps the old quantity	Simulator > Scenarios > S5: tick 'Reset first', 0 s between steps, Run S5. In the worklist click the top-bar pill.	New flags on 76108: R05-76108, R07-WO-UBD-20260912-001.	run: Run Scenario 5: MIPS re-plans the work order; ODI keeps the old quantity FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: MIPS saves the bigger work order (+1,365). (Published event #50 | worklist pill: 1 row changed by others · Update | new flags: R05-76108 (1,365 u $6,402), R07...	**PASS**
I6	S6: A WMS cycle count changes on hand only	Simulator > Scenarios > S6: tick 'Reset first', 0 s between steps, Run S6. In the worklist click the top-bar pill.	New flags on 76108: R10-INV-0001.	run: Run Scenario 6: A WMS cycle count changes on hand only FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: WMS books the count against on hand (-100). (Published event #509 (source change | worklist pill: 1 row changed by others · Update | new flags: R10-INV-0001 (100 u $469)	**PASS**
I7	S7: The line runs late: wrapping ends after the ship dates	Simulator > Scenarios > S7: tick 'Reset first', 0 s between steps, Run S7. In the worklist click the top-bar pill.	New flags on 76108: R12-WO-UBD-20260912-001.	run: Run Scenario 7: The line runs late: wrapping ends after the ship dates FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: ODI records the late wrap on every row of the work order. (Publis | worklist pill: 1 row changed by others · Update | new flags: R12-WO-UBD-20260912-001 (4,664 ...	**PASS**
I8	S8: The work order is deleted in MIPS	Simulator > Scenarios > S8: tick 'Reset first', 0 s between steps, Run S8. In the worklist click the top-bar pill.	New flags on 76108: R06-76108, R07-WO-UBD-20260912-001.	run: Run Scenario 8: The work order is deleted in MIPS FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: MIPS deletes the work order. (Published event #509 (source change S509).) 20:00:00 Don | worklist pill: 1 row changed by others · Update | new flags: R06-76108 (48,635 u $228,098), ...	**PASS**
I9	S9: ODI loses the work-order link on a demand line	Simulator > Scenarios > S9: tick 'Reset first', 0 s between steps, Run S9. In the worklist click the top-bar pill.	New flags on 76108: R08-ODI-0002.	run: Run Scenario 9: ODI loses the work-order link on a demand line FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: ODI rewrites the Albany row without its work order. (Published event #509 | worklist pill: 1 row changed by others · Update | new flags: R08-ODI-0002 (7,200 u $33,768)	**PASS**

ID	Check	Steps	Expected	Observed	Result
I10	S10: An order arrives with a mistyped BUSS item code	Simulator > Scenarios > S10: tick 'Reset first', 0 s between steps, Run S10. In the worklist click the top-bar pill.	New flags on 76108: R13-76108.	run: Run Scenario 10: An order arrives with a mistyped BUSS item code FINISHED 20:00:00 Reset to the starting day. 20:00:00 Step 1/1: BUSS sends the order with the wrong code. (Published event #509 (source | worklist pill: 6 rows changed by others · Update | new flags: R13-76108 (0 u $0)	PASS
I11	Correct Scenario 1 end to end through write-back	After S1, Pat approves the proposed corrections on 76108 in turn: R02 (ODI), then R04 (MIPS demand), then R05 (MIPS work order), clicking Update after each.	Each correction is applied in the source system and confirmed by the feed; the next proposal is recomputed from the corrected data; after the third, 76108 reconciles and leaves the queue.	R02 -> ODI: open 76108 flags ['R04-76108', 'R05-76108']; R04 -> MIPS: open 76108 flags ['R05-76108']; R05 -> MIPS: open 76108 flags none	PASS
P11	S11: R14 in the worklist	Run S11 with Reset first; click Update; open 76108 and its evidence.	Only R14-2263245-1 is raised on 76108, High, with the robust z-score, the history and the accountability.	76108 flags: ['R14-2263245-1']; flag text: HIGH R14 Order quantity unusual for this customer 64,800 units · $303,912 From the ML layer · robust z-score against this item's history BUSS order 2263245 from ALBANY NY (PROSPECT RD) BAKERY is for 72,000 units of 76108, against a usual 7,200 (median of it...	PASS
P12	S12: R15 in the worklist	Run S12 with Reset first; click Update; open 76108 and its evidence.	Only R15-FC001 is raised on 76108, High, with the robust z-score, the history and the accountability.	76108 flags: ['R15-FC001']; flag text: HIGH R15 Forecast out of line with its history 41,769 units · $195,897 From the ML layer · robust z-score against this item's history Oracle forecast FC001 for 76108 is 83,538, against a usual 41,769 (median of the last 8 sales days; robust z = +19.3). ODI and	PASS
P13	S13: R16 in the worklist	Run S13 with Reset first; click Update; open 76108 and its evidence.	Only R16-WO-UBD-20260912-001 is raised on 76108, High, with the robust z-score, the history and the accountability.	76108 flags: ['R16-WO-UBD-20260912-001']; flag text: HIGH R16 Wrap duration unusual for this item 0 units · — No retail value From the ML layer · robust z-score against this item's history Work order WO-UBD-20260912-001 (Line 1 Shift G) took 9.7 hours to wrap 48,635 units, against a usual 4.6 hours ...	PASS
P14	S14: R17 in the worklist	Run S14 with Reset first; click Update; open 76108 and its evidence.	Only R17-INV-0001 is raised on 76108, High, with the robust z-score, the history and the accountability.	76108 flags: ['R17-INV-0001']; flag text: HIGH R17 Unexplained change in WMS stock 3,000 units · $14,070 From the ML layer · robust z-score against this item's history WMS on hand for 76108 at FG.1.1.1 went from 7,473 to 4,473 (-3,000) at 09/11 20:00. Its last 8 changes moved it by 149 units either ...	PASS
J1	Export the source data now	Click Excel, SQLite and Postgres under 'Export now'.	Each downloads the current rows: an Excel workbook (one sheet per source, the position and the flags), a SQLite database with the v_sku_position view, and a PostgreSQL script.	Excel: Source_Data_Now-20260911-2104.xlsx (50,913 bytes, valid True); SQLite: source_data_now-20260911-2105.sqlite (159,744 bytes, valid True); Postgres: source_data_now_postgres-20260911-2105.sql (232,188 bytes, valid True); v_sku_position rows: 6	PASS
K1	Reset to the starting day from the simulator	In the simulator click 'Reset to starting day' and confirm (done after phase 3). Look at the worklist.	The starting day is published again and the worklist shows the eight seed flags open, as after the worklist's own Reset simulation.	DEFECT. The source data is reset, but the worklist keeps the decisions made before it: on the fresh starting day 4 of 8 flags open as 'Accepted - awaiting correction' (R05-950513, R02-2263255-1, R10-INV-0003, R11-WO-UBD-20260912-002), including R05 and R02 whose corrections had already been confirme...	FAIL
K2	Reset simulation from the worklist	Pat clicks 'Reset simulation' in the worklist's top bar and confirms. Sam has the worklist open.	The starting day is restored and decisions, claims, acknowledgments and notes are cleared. Sam's top bar says 'Simulation reset to the starting day · Reload'; after Reload the eight seed flags are open.	Sam's notice: Simulation reset to the starting day · Reload; open flags: 8; decided: 0	PASS
L1	Sign out of both apps	Click Sign out in the worklist and in the simulator.	Each returns to its sign-in page.	worklist: /signin; simulator: /signin	PASS
M1	Phone and tablet layouts	Open the worklist at 390 px (phone) and 1024 px (tablet).	Phone: the queue fills the screen and a SKU opens as a full-screen detail with a Back link; top bar on two lines. Tablet: the detail slides over the queue; account and assistant become icons.	phone and tablet screens captured; no horizontal scroll: True	PASS

ID	Check	Steps	Expected	Observed	Result
N1	The Reconciliation API	Call each endpoint with a signed-in session (and one without).	Health is public; every /api endpoint returns JSON for a signed-in user and refuses an anonymous caller.	GET /healthz: {"status":"ok","version":"0.5.0","source":"events"}; GET /api/flags: 9 open flags; GET /api/positions/950513: shortfall 3700.0, flags ['R05-950513', 'R11-WO-UBD-20260912-002']; GET /api/briefing/day: source Rule engine (offline - set the OCI_* credentials in .env for AI narratives), 5 …	**PASS**
O1	Version	python -m recon_agent --version	Prints the version.	recon_agent 0.5.0	**PASS**
O2	List accounts	python -m recon_agent users list	Lists each account with its name and role.	olly planner Olly Operator pat planner Pat Planner sam supervisor Sam Supervisor mona manager Mona Manager	**PASS**
O3	Event log commands	events seed --reset; status; show; set; tail; scenario list; scenario run S6 --reset; clock --advance 15m; export; entities (on a separate log).	Each command succeeds: the log is seeded, rows are shown and changed, scenarios run, the clock moves and the export files are written.	events seed --reset: exit 0, Published 508 synthetic events for sales date Sat 2026-09-12. Clock: Fri 2026-09-11 20:00:; events status: exit 0, Log C:\Users\Ajinkya Bhosale\AppData\Local\Temp\claude\f96f9066cecdf7e7\8bce2c6a-e3c4-4e34; events show wms.inventory: exit 0, Inventory ID Item Number Item…	**PASS**
O4	Reconcile the client workbook to a report	RECON_SOURCE=excel python -m recon_agent run --offline --out-dir …	Reads the business workbook under data/, finds 76108 reconciled (no flags), and writes the planner workbook and the HTML dashboard.	exit 0; Sales date 2026-09-12: 1 SKUs, 0 exceptions ({}), $0 exposure demand 56,108 · shortfall 48,635 · work orders 48,635 · over-production 0 · under-production 0 uni; files ['recon_2026-09-12.html', 'recon_2026-09-12.json', 'recon_2026-09-12.xlsx']	**PASS**
O5	The worklist on the business workbook (not live)	Start the worklist with RECON_SOURCE=excel and the client workbook under data/.	The top bar says 'Source workbook, not live' with the load time; 76108 reconciles, so the queue explains that every SKU reconciles; no Reset simulation button.	Exception worklist Source workbook, not live MIPS · ODI · WMS for SAT 09·12 Sales day Saturday 12 September 2026 loaded 18:44 My claims 0 Ask PP Pat Planner Pla | Open reconciliation flags Shift briefing From the rules · AI text on the way · 18:44 Read briefing 0 reconciliation exceptions for sales …	**PASS**
O6	Ask from the command line	python -m recon_agent ask "Is 76108 reconciled …?"	Prints an answer grounded in the workbook's figures.	Yes, SKU 76108 (BP WHITE HOT DOG BUNS 8P 14Z) is reconciled. The work order is WO-UBD-20260912-001 for 48,635 units. This is correct: total demand is 56,108 units, and WMS has 7,000 units available, leaving a shortfall of 49,108 to produce.	**PASS**

The automated tests (141, all passed)

Run with `python -m unittest discover -s tests -v`. Grouped by test file and class.

test_accountability · AttributionTest (5)	every flag has a complete record; kpis total the flags by origin; quantities are written unsigned; the scenarios are pinned on the right system; the seeded problems are pinned on the right system
test_accountability · AuditLogTest (4)	a failed call is logged too; a narration is logged and its entries point at the log; an assistant answer is logged with its tool calls; text that cannot be logged is not shown
test_accountability · CardTest (1)	the card shows accountability and the ai trail
test_accountability · FigureCheckTest (5)	a linked flags figures count as known; an invented figure is reported; bad citations are dropped and recorded; codes dates and small counts are not figures; figures from the engine pass
test_anomaly · DetectorTest (3)	each ml scenario raises one high flag that explains itself; ordinary variation stays quiet; the threshold is configurable
test_anomaly · HistoryTest (3)	a single day raises no ml flags; history never enters reconciliation; wms history is one snapshot per change of on hand
test_anomaly · RobustZTest (3)	a flat history does not divide by zero; nothing is judged without enough history; the median scores zero and one outlier cannot hide the next
test_data_model · EquationTest (5)	an item with only a forecast is planning until it has orders; an odi row served from stock belongs to its order; each source moves only its own term; mips that ignores allocated stock is named; stock that covers the demand needs no work order
test_data_model · LedgerTest (1)	the card draws the equation
test_data_model · ProposalTest (1)	several work orders need a person
test_events · BusinessWorkbookTest (2)	76108 reconciles as the business team worked it; the seed carries 76108 value for value
test_events · ClockTest (2)	controls; durations
test_events · ContractTest (5)	bool parsing; full typed image; one entity per workbook sheet; rejects unknown and missing columns; round trip keeps values
test_events · LogTest (4)	append only; insert update delete; rejects invalid changes atomically; reset starts a new log
test_events · ProjectorTest (2)	arrival order does not change the result; incremental and reset
test_events · SeedTest (6)	another sales date moves every date; empty log runs; prices and stock served lines; seed needs an empty log; seed raises exactly the designed flags; the control sku is the workbook example
test_events · StoreTest (2)	ack sla follows the simulated clock; worklist reacts to an event
test_events · WorkbookFeedTest (3)	the price column before 10 nov is still read; the workbook and the event feed reconcile the same; the workbook is found anywhere under the data directory
test_live · ChangeReachesScreensTest (4)	a change to a planning only item changes nothing on screen; ai text is asked again when the wording changes but not the units; an edit that moves no flag still reaches the screen; the first row of an empty worklist is announced
test_live · EveryOdiRowTest (3)	a stale row that is not the first is found and counted; batch rows that disagree are named; the lawrap of the late rows is reported
test_live · ExportTest (3)	the database view recomputes the position; the simulator serves the export to operators only; the workbook carries the current rows and flags
test_live · LiveRedrawTest (4)	a feed change waits for update; ai text redraws the open detail; an open row that clears says so; the assistant suggests questions about the current queue
test_live · LiveScenarioTest (3)	a scenario that cannot run says why; changes are relative to the current row; the preview follows the data
test_scenarios · FaultTest (7)	a later full image heals a dropped update; delay holds the event then delivers; drop loses the event until redelivered; duplicates are ignored by the consumer; flush and removal; reordered events keep the newest image; seed bypasses faults and reset clears them
test_scenarios · ScenarioLibraryTest (3)	a scenario needs the starting rows; a step lands in one commit; every scenario raises exactly its flags
test_security · AccountsTest (3)	demo accounts cover every role; random demo passwords when none is given; rules for new accounts
test_security · PrimitivesTest (4)	atomic json; passwords; rate limiter window; signed tokens
test_security · SimulatorSignInTest (4)	corrections token is enforced when set; crawler and browser files; operator cookie cannot be forged; operator signs in with an account
test_security · WorklistSignInTest (17)	anonymous is sent to sign in; crawler and browser files; cross site posts are refused; error fragments are escaped; expired session is refused; forged and plain cookies are ignored; gateway headers are not trusted; health and docs; name and role come from the account; password change and removal end sessions; repeated failures lock the username; role cannot be chosen at sign in; role change applies at once; security headers; sign in never redirects off site; simulator cookie does not open the worklist; wrong password is refused
test_simulator · NarrationTest (1)	debounced and cached
test_simulator · RealtimeTest (2)	stream announces a new version; worklist commits and latency is measured
test_simulator · SimulatorTest (10)	bad value is refused and kept on screen; clock controls; edit publishes the changed columns only; insert copy and delete; next key; poll refreshes the table only when the log moved; requires an operator; reset reseeds; stale form is refused; unchanged form publishes nothing

test_worklist_ui · WorklistTruthTest (6)	a decision moves the money and the counts; an empty view is never an all clear; every accept steps back while a high flag waits; handed over and the since line; one unit is singular; the top row is not current until its detail is open
test_writeback · ResetSimulationTest (1)	reset from the worklist returns to the starting day
test_writeback · WriteBackTest (9)	a reset retires earlier corrections; approved correction is confirmed by the feed; endpoint down then retry; inserts deletes and links; only planners and managers approve; restoring a deleted work order clears the odi link; slow integration keeps it sent until the event arrives; stale row is refused; the whole chain is corrected system by system

141 test names listed; the run reported 'Ran 141 tests ... OK'.

15. Glossary

Term	Meaning
SKU	An item, such as 76108 BP WHITE HOT DOG BUNS 8P 14Z.
Flag	A mismatch between systems found by one of the 13 rules.
Reconciles	All systems agree for the SKU: it has no flags.
Allocated / available	WMS stock already reserved for orders / free to use.
Shortfall	What must still be made: demand less allocated and available stock.
Severity	High, Medium or Low. High flags must be acknowledged within 15 minutes.
$ at risk	Units × the BUSS retail price. A SKU counts once, by its largest flag.
Priority	$ at risk × severity × rule confidence: the queue order.
Claim	Telling others you are working on a SKU.
Acknowledge	Confirming you have seen a High flag.
Shift note	A note on a SKU for the next shift.
Correction / write-back	The change the agent sends back to the source system once approved.
Oracle Integration (OIC)	The middleware that receives corrections; in the demo, the simulator.
Event / feed	A change in a source system, delivered to the worklist by the integration.
Plant time	The simulated clock of the demo plant.
Scenario	A scripted change that breaks one step of the flow.
ML layer	Anomaly detection (R14-R17) that compares today with each item's own history.
Robust z-score	How far a value is from its history's median, in units of the median absolute deviation. 3.5 or more is unusual; 7 or more is High.